

Dynamic Lorenz Dominance

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Abstract

This paper extends Lorenz dominance from income distributions to transitions, providing an ordinal, index-free criterion for inequality reduction. Dynamic Lorenz dominance compares centered cumulative rank-based increments in two dimensions: absolute, using income changes, and relative, using log-income changes. On fixed-mean, fixed-population domains, Symmetry, a Dynamic Pigou–Dalton Principle, and Centered Baseline Independence characterize dynamic Lorenz dominance, which is equivalent to decomposing transition profiles into progressive transfers. The absolute dimension recovers Lorenz dominance. The relative dimension applies Lorenz-type cumulative dominance to centered log-increment profiles and, against a null transition, yields centered log-Lorenz dominance. Extensions cover variable means and populations through dimension-specific invariance and a Dynamic Population Principle.

Keywords: static Lorenz dominance, dynamic Lorenz dominance, dynamic inequality.

JEL classifications: D63, D71.

1 Introduction

The Lorenz criterion is the standard benchmark for unambiguous inequality comparisons across income distributions. It is characterized by the Pigou–Dalton transfer principle (Pigou, 1912; Dalton, 1920) through the majorization theorem of Hardy, Littlewood, and Pólya (cf. Hardy et al., 1952; Atkinson, 1970; Rothschild and Stiglitz, 1973; Dasgupta et al., 1973; Fields and Fei, 1978). While particular Pigou–Dalton-consistent indices may disagree, they agree whenever Lorenz dominance applies. Yet economists are often interested not only in comparing inequality across distributions, but also in evaluating how inequality changes between them. This dynamic perspective is particularly important in policy analysis.

Tax reforms, transfer programs, and episodes of economic growth are evaluated through the income transitions they generate, not solely through the inequality of their endpoints. Policymakers therefore require a way to compare the inequality-reducing content of competing reforms without committing to a particular inequality index. Current practice often relies on before-and-after comparisons of cardinal inequality measures, which may conflate redistribution with aggregate growth. An ordinal, index-free criterion for dynamic inequality is therefore desirable.

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This paper develops such a criterion by extending the static inequality framework to income transitions. The primitive object is a transition from an initial distribution $\mathbf{x}$ to a final distribution $\mathbf{y}$. A dynamic inequality pre-order $\succsim$ ranks such transitions:

$$(\mathbf{x}', \mathbf{y}') \succsim (\mathbf{x}, \mathbf{y})$$

means that the first transition represents a weakly larger reduction in inequality. Static comparisons are embedded as comparisons with the null transition $(\mathbf{x}, \mathbf{x})$: evaluating

$$(\mathbf{x}, \mathbf{y}) \succsim (\mathbf{x}, \mathbf{x})$$

recovers the comparison between $\mathbf{x}$ and $\mathbf{y}$.

The framework distinguishes two dimensions. The absolute dimension ($K = A$) uses raw income increments; the relative dimension ($K = R$) uses logarithmic increments, separating uniform scale growth from redistribution.

We begin on a fixed-mean, fixed-population domain by imposing three axioms: Symmetry (**S**), the K -Dynamic Pigou–Dalton Principle (**K-DPDP**), and Centered K -Baseline Independence (**K-CBI**). Symmetry requires evaluations to be anonymous in both periods. **K-DPDP** extends Pigou–Dalton to transitions: if one transition’s increment profile is obtained from another by adding progressive enhancements, it must be strictly preferred. **K-CBI** requires dynamic evaluations to depend only on centered rank-based K -increment profiles, factoring out both the initial state of the economy and any aggregate economic growth.

We then define K -Dynamic Lorenz dominance ($\succsim_{DL}^K$). For each rank k , the criterion compares the cumulative K -increments received by the bottom k ranks with their population share k/n of aggregate K -growth. Thus, a transition is more equalizing if every bottom segment receives at least its fair share of the gain.

The main fixed-domain result, **Theorem 1**, shows that a dynamic inequality pre-order satisfies **S**, **K-DPDP**, and **K-CBI** if and only if it is consistent with $\succsim_{DL}^K$. The representation step applies classical majorization geometry to the centered difference between rank-based increment profiles: K -dynamic Lorenz dominance is equivalent to the decomposability of rank-based increment gaps into the progressive enhancements required by **K-DPDP** (**Lemma 2**). Once the relevant increment profiles and their neutral components have been identified, the equivalence follows from the standard cumulative-sum decomposition. The substantive contribution is to identify the dynamic objects and baseline-neutrality conditions under which this familiar geometry represents inequality reduction across transitions. Against the null transition, the absolute dimension recovers standard Lorenz dominance. The relative dimension yields centered log-Lorenz dominance, which applies ordinary majorization to log incomes after removing their mean-log, or geometric-mean, component (**Hardy et al., 1952; Marshall and Olkin, 1979; Marshall et al., 2011**). Logarithmic majorization and its multiplicative transfer representation have antecedents in **Aboudi and Thon (2003)**, building on the proportional-transfer analysis of **Fleurbaey and Michel (2001)**. Our static result has a different purpose: **Theorem 2** characterizes centered log-Lorenz dominance as the minimal scale-neutral inequality ordering generated by Symmetry and the Centered Proportional Transfer Principle on a fixed-arithmetic-mean domain. Its principal role in this paper is to identify and axiomatize the static restriction of the relative dynamic theory; the principal contribution is the extension and axiomatic interpretation of this centered Lorenz geometry for comparisons of transitions rather than states.

The framework is then extended beyond fixed means. For variable total incomes, the relevant invariance differs by dimension. The absolute dimension requires Dynamic Translation Invariance

(DTI), while the relative dimension requires Dynamic Scale Invariance (DSI). Both are implied by the corresponding global version of *K-CBI* (Proposition 2). This yields the variable-mean characterizations in Theorem 3 and Theorem 4. The final extension allows population sizes to vary. We impose the Dynamic Population Principle (DPP), a dynamic analogue of Dalton’s replication principle, and obtain the unrestricted characterizations in Theorem 5 and Theorem 6.

The paper is related to several strands of literature. The income mobility literature (e.g., Shorrocks, 1978; Fields and Ok, 1999) typically tracks individual trajectories across periods. Cowell (1985) develops an axiomatic theory of distributional change, encompassing mobility and horizontal inequity, and characterizes cardinal indices defined on matched initial and final incomes. Plotnick (1982) obtains an ordinal, Lorenz-type criterion for comparing horizontal inequity across redistributions. Both approaches retain individual-level information across periods: Cowell’s measures depend on matched income pairs, while Plotnick’s criterion evaluates departures from individuals’ initial ranks. Our framework instead imposes anonymity separately in each period. It compares transitions between marginal distributions and therefore abstracts from reranking and exchange mobility.

Other related work studies distributional progressivity without requiring individual trajectories. Jenkins and Van Kerm (2006) decompose changes in cardinal inequality indices into progressivity and reranking components. Ravallion and Chen (2003) and Son (2004) use growth-incidence curves to study pro-poor growth and poverty reduction. Ray and Genicot (2023) axiomatize upward mobility as pro-poor growth; Growth Progressivity leads to a cardinal measure linear in growth rates with weights decreasing in baseline income. Our object is instead an ordinal comparison of the inequality-reducing content of transitions, based on centered rank-based increment profiles.

The closest paper is Bossert and Dutta (2019). They study cardinal welfare change and show that second-order dominance of rank-ordered income differences is equivalent to unanimous welfare-change comparisons within the generalized-Gini class. On a fixed-mean domain, this dominance condition coincides with our absolute dynamic Lorenz dominance. The approaches differ in object, scope, and axiomatic foundation. Bossert and Dutta impose welfare difference compatibility and retain aggregate growth as part of welfare change. We rank inequality change directly and, when means vary, remove aggregate growth by centering increments. Moreover, our relative dimension uses centered log-increments and yields a proportional, scale-invariant dominance criterion with no counterpart in their raw-difference framework. Finally, our characterization is based on Symmetry, the Dynamic Pigou–Dalton Principle, and Centered Baseline Independence rather than on a cardinal representation by differences of generalized-Gini welfare levels.

The relative branch also connects to transformed-majorization and proportional-transfer results. Aboudi and Thon (2003) characterize weak supermajorization of ordered log incomes through multiplicative T -transforms, coordinatewise increases, and unanimity over additive welfare functions that are more concave than the logarithm. Their order retains changes in the geometric-mean level and is therefore the logarithmic analogue of generalized Lorenz welfare dominance. Our centered criterion removes that level component and is the logarithmic analogue of Lorenz inequality dominance. Fleurbaey and Michel (2001) characterize additive welfare functions satisfying proportional and proportional ex post transfer principles. Their elementary proportional ex post transfer is not itself a zero-sum Pigou–Dalton transfer in log income. Aboudi and Thon (2003) show, however, that sequences of increasingly small proportional ex post transfers converge to the product-preserving multiplicative transformations generated by Pigou–Dalton transfers in log-income space. Neither paper compares transitions through centered rank-based log-increment profiles, which is the

object of the relative dynamic theory developed here.

The paper also relates to the transfer-sensitivity analysis of [Shorrocks and Foster \(1987\)](#). Their criterion strengthens static Pigou–Dalton comparisons by favoring physical transfers lower in the income distribution. Our absolute dynamic criterion instead compares the magnitude and rank reach of endpoint increment profiles. The relative criterion evaluates centered log-increments and therefore measures proportional rather than physical transfer intensity.

The rest of the paper is organized as follows. [Section 2](#) develops the fixed-mean, fixed-population model and proves the main characterization. [Section 3](#) extends the analysis to variable total incomes. [Section 4](#) extends the framework to varying population sizes. [Section 5](#) concludes.

2 Distributions with the same total income and population size

2.1 Basic setup

We represent an *income distribution* for a population of size n as an n -dimensional vector of nonnegative incomes $\hat{x} = (\hat{x}_1, \dots, \hat{x}_n) \in \mathbb{R}_+^n$, where $\hat{x}_i$ denotes the income of individual i .

In this section, we restrict our attention to the domain of distributions sharing a fixed population size $n \in \mathbb{N}$ and a strictly positive fixed mean income μ . Let $\Omega_{(n,\mu)}$ denote the space of all such unordered distributions.

To analyze inequality over time, we consider *transitions* between distributions, represented as pairs $(\hat{x}, \hat{y}) \in \Omega_{(n,\mu)} \times \Omega_{(n,\mu)}$ spanning two periods. Our goal is to define a *dynamic inequality pre-order* $\succsim$ on the space of transitions to capture the intensity of changes in inequality between periods: $(\hat{x}', \hat{y}') \succsim (\hat{x}, \hat{y})$ means that the reduction in inequality from $\hat{x}'$ to $\hat{y}'$ is weakly larger than the reduction from $\hat{x}$ to $\hat{y}$.

Such a measure naturally encompasses the static case, represented by the null transition $(\hat{x}, \hat{x})$, where evaluating $(\hat{x}, \hat{y}) \succsim (\hat{x}, \hat{x})$ collapses to the standard static comparison between $\hat{x}$ and $\hat{y}$. Throughout the sequel, $\succsim$ is assumed to be reflexive and transitive on the space of transitions.

2.2 Symmetry

Because individuals differ only by income, any dynamic inequality evaluation should be invariant under the relabeling of individuals within either period. We formalize this ethical premise via the standard anonymity postulate.

Symmetry (S). If $\hat{x}'$ and $\hat{y}'$ result from $\hat{x}$ and $\hat{y}$ by a permutation of individuals' incomes, then $(\hat{x}', \hat{y}') \sim (\hat{x}, \hat{y})$.

By imposing **S**, the pre-order evaluates distributions anonymously. Consequently, there is no loss of generality in evaluating these profiles exclusively via their sorted order statistics. For any $\hat{x} \in \Omega_{(n,\mu)}$, let $\mathbf{x} = (x_1, \dots, x_n)$ denote its rank-ordered counterpart, where $x_1 \leq \dots \leq x_n$ and x_i represents the income of the individual occupying rank i in the distribution. The term *rank* refers to this position in the sorted sequence, not to an individual's fixed label.

Let $\mathcal{D}_{(n,\mu)}$ denote the space of all such rank-ordered distributions. Most of the analysis is restricted to the ordered subspace $\mathcal{D}_{(n,\mu)} \times \mathcal{D}_{(n,\mu)}$, operating on the sorted transitions $(\mathbf{x}, \mathbf{y})$.

2.3 Transfers and elementary decompositions

Understanding how a single transfer alters an income distribution is central to the analysis of rank-based inequality measures. A classical transfer between two individuals may induce rank reshuffling, altering the global order of the distribution. In what follows, we establish that any such transition can be decomposed into a sequence of globally rank-preserving transfers.

This decomposition allows us to describe dynamic changes in inequality in terms of ordered income shifts, without tracking individual identities. Any observed transition over time can thus be interpreted as the cumulative effect of these rank-preserving steps. The formal construction is relegated to [Appendix A.1](#).

2.3.1 Transfer notation and setup

Definition 1 (elementary transfer). An *elementary transfer* from rank d to rank r of amount $t > 0$ applied to distribution $\mathbf{x}$ is a transformation $\mathbf{x} \mapsto \mathbf{y}$ defined as follows.

Let the individual occupying rank r in $\mathbf{x}$ receive t units, and let the individual occupying rank d in $\mathbf{x}$ give t units. All other individuals' incomes remain unchanged. The resulting distribution is $\mathbf{y}$.

The transfer must satisfy the relative rank-preservation condition: after the transfer, the relative income ranking between the donor and recipient is preserved. That is, if a and b denote the individuals at ranks r and d respectively in $\mathbf{x}$, then $y_a \leq y_b$ when $r < d$, and $y_a \geq y_b$ when $r > d$.

The transfer is called *progressive* if $r < d$ (income flows from higher to lower rank), and *regressive* if $r > d$ (income flows from lower to higher rank).

The definition above requires that the relative income ranking between the donor and recipient be preserved. However, an elementary transfer permits global rank reshuffling, as the donor or recipient may leapfrog intermediate individuals. We identify a restricted subclass of transfers that induce no such reshuffling.

Definition 2 (globally rank-preserving transfer). An elementary transfer $\mathbf{x} \mapsto \mathbf{y}$ applied to an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ is *globally rank-preserving* if the post-transfer distribution remains in the ordered domain, i.e., $\mathbf{y} \in \mathcal{D}_{(n,\mu)}$.

The distinction between these two definitions is bridged by a fundamental decomposition property. While a standard elementary transfer induces complex rank reshuffling, it can always be reduced to a sequence of globally rank-preserving transfers.

Proposition 1. Any elementary transfer applied to an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ can be decomposed into a finite sequence of globally rank-preserving transfers. Furthermore, if the elementary transfer is progressive (respectively, regressive), its decomposition consists exclusively of globally rank-preserving progressive (respectively, regressive) transfers.

The proof of [Proposition 1](#) is provided in [Appendix A.1](#).

To see how these globally rank-preserving transfers are encoded in the rank-based increment vector, consider a 4-person economy with an initial distribution $\mathbf{x} = (10, 20, 50, 70)$. Suppose we execute a single elementary progressive transfer of 25 units from the richest individual to the poorest.

The physical transfer yields the unordered target distribution $(35, 20, 50, 45)$. Re-sorting this array yields the final ordered target $\mathbf{y} = (20, 35, 45, 50)$. We can now compute the absolute

rank-based increment vector for this transition:

$$\boldsymbol{\tau}^A = \mathbf{y} - \mathbf{x} = (10, 15, -5, -20).$$

Because the elementary transfer induced rank reshuffling (the recipient leapfrogged rank 2, and the donor dropped below rank 3), the increment vector is spread across four ranks. This vector encodes the globally rank-preserving transfers guaranteed by [Proposition 1](#).

The positive entries dictate how much income the moving recipient absorbs at each rank tier, while the negative entries dictate how much the moving donor expends. By matching the lowest-rank gains to the highest-rank losses, we reconstruct the decomposition:

- **Step 1:** Rank 1 requires a gain of 10. We fund this from the 20-unit deficit at rank 4, yielding the rank-preserving progressive transfer $10(\mathbf{e}_1 - \mathbf{e}_4)$.
- **Step 2:** Rank 2 requires a gain of 15. We exhaust the remaining 10-unit deficit at rank 4, yielding the rank-preserving progressive transfer $10(\mathbf{e}_2 - \mathbf{e}_4)$.
- **Step 3:** Rank 2 still requires a gain of 5. We fund this from the remaining 5-unit deficit at rank 3, yielding the final rank-preserving progressive transfer $5(\mathbf{e}_2 - \mathbf{e}_3)$.

Thus, the rank-based increment vector generates the transfer decomposition.

2.3.2 Rank-based increments

For any transition from an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ to a final distribution $\mathbf{y} \in \mathcal{D}_{(n,\mu)}$, we characterize the distributional change via the *rank-based increments*. We evaluate the transition using:

- **Absolute increments:** $\tau_i^A = y_i - x_i$, representing the raw dollar shift.
- **Relative increments:** $\tau_i^R = \ln y_i - \ln x_i$, representing the logarithmic proportional shift. Because the logarithm is undefined at zero, we restrict the domain for the relative dimension to strictly positive incomes, denoted $\mathcal{D}_{(n,\mu)}^+ = \{\mathbf{x} \in \mathcal{D}_{(n,\mu)} : x_1 > 0\}$.

When the transition is a globally rank-preserving transfer of amount $t > 0$ from a donor at rank d to a recipient at rank r , the absolute increment vector $\boldsymbol{\tau}^A$ takes a particularly simple form. Because no rank reshuffling occurs, incomes outside these two specific ranks remain unchanged:

$$\tau_r^A = t, \tau_d^A = -t, \text{ and } \tau_i^A = 0, \quad \text{for all } i \notin \{r, d\}.$$

The relative increment vector $\boldsymbol{\tau}^R$ similarly isolates changes to ranks r and d , yielding strictly positive τ_r^R and strictly negative τ_d^R , though their magnitudes depend on the baseline incomes.

Remark 1. The relative dimension uses logarithmic differences $\tau_i^R = \ln y_i - \ln x_i$ rather than standard percentage changes. Log differences are well suited to this purpose ([Törnqvist et al., 1985](#)): doubling and halving generate changes of equal magnitude and opposite sign; changes are additive over time; and uniform proportional growth becomes an additive constant in log-space, allowing centering to separate aggregate scale from redistribution. The use of log income as a primitive socioeconomic coordinate also has precedent in [Esteban and Ray \(1994\)](#), who motivate it by the relevance and symmetry of proportional income differences. Their application is cross-sectional polarization, whereas ours uses log differences to evaluate intertemporal changes across ranks.

2.4 Enhancements and transfer intensity

The previous subsection established that globally rank-preserving transfers possess a simple increment structure. To formally evaluate dynamic inequality, we must determine when one transition is “more equalizing” than another. We do this by defining *elementary enhancements* that unambiguously improve a transition’s redistributive power.

We define these enhancements across two parallel dimensions: an *absolute* dimension, which evaluates raw income shifts, and a *relative* one, which evaluates proportional sacrifices and gains via logarithmic shifts. As we will demonstrate, distinguishing between absolute and relative enhancements overcomes the linearity constraints intrinsic to purely absolute frameworks (see [Bossert and Dutta, 2019](#)).

Definition 3 (*K*-dominance). Let $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{x}', \mathbf{y}')$ be two transitions with equal total income and population size, generating rank-based increment vectors $\boldsymbol{\tau}^K$ and $\boldsymbol{\tau}'^K$. We say the transition $(\mathbf{x}', \mathbf{y}')$ *K-dominates* $(\mathbf{x}, \mathbf{y})$ if $\boldsymbol{\tau}'^K$ can be obtained from $\boldsymbol{\tau}^K$ by superimposing:

- **Absolute dimension** ($K = A$): A finite, nonempty sequence of *progressive increment transfers* of the form $a(\mathbf{e}_r - \mathbf{e}_d)$ with $a > 0$ and $r < d \leq n$. This represents a pure, zero-sum exchange in raw dollars.
- **Relative dimension** ($K = R$): A finite, nonempty sequence of *progressive increment log-transfers* $a(\mathbf{e}_r - \mathbf{e}_d)$ (where $a > 0$ and $r < d \leq n$) and a single *inequality-neutral uniform shift* $c\mathbf{1}$ (where $c \in \mathbb{R}$). This represents a pure compression of relative income shares, explicitly separating pure redistribution from aggregate proportional growth.

The two dimensions capture different notions of progressive intensity. In the absolute dimension, superimposing $a(\mathbf{e}_r - \mathbf{e}_d)$ increases the raw amount redistributed, its rank reach, or both. These enhancements compare realized increment profiles rather than physical transfers executed on a particular baseline; the distinction is discussed in [Section 2.6.2](#).

In the relative dimension, the same zero-sum operation is applied to log-increments and therefore represents a progressive proportional redistribution. The uniform term $c\mathbf{1}$ absorbs the inequality-neutral scale component, leaving the centered log-profile to capture pure relative progressivity.

2.5 The dynamic Pigou–Dalton framework: a unified approach

Extending the classical Pigou–Dalton transfer principle to a dynamic setting requires a conceptual shift. The static principle evaluates *states*, declaring that transferring income from a richer person to a poorer person yields a more equal society. However, a dynamic pre-order evaluates *transitions*, measuring the magnitude of equalization over time.

Because real-world reforms often entail complex redistributions and rank reshuffling, evaluating dynamic inequality requires assessing a transition’s net profile. If one policy’s redistributive effect can be constructed from a baseline by superimposing progressive updates, it is unambiguously more equalizing. We formalize this logic into a single dynamic axiom based on the profiles defined in [Definition 3](#).

K-Dynamic Pigou–Dalton Principle (K-DPDP). For $K \in \{A, R\}$, a dynamic inequality pre-order $\succsim$ satisfies the **K-DPDP** if, for any two transitions $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{x}', \mathbf{y}')$ whose distributions belong to

the relevant domain ($\mathcal{D}_{(n,\mu)}$ when $K = A$, and $\mathcal{D}_{(n,\mu)}^+$ when $K = R$), if $(\mathbf{x}', \mathbf{y}')$ K -dominates $(\mathbf{x}, \mathbf{y})$, then $(\mathbf{x}', \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$.

K -DPDP serves as a dynamic extension of the classical Pigou–Dalton logic. It operates via cumulative progressivity and ranks isolated redistributions by both their *intensity* and *reach*. A globally rank-preserving progressive transfer bridges a wider rank gap or redistributes a larger raw increment (Figure 1) precisely when its profile can be constructed by superimposing additional progressive enhancements onto the weaker baseline transfer. By evaluating these abstract superimpositions, the **K -DPDP** enforces the rule that larger and wider transfers yield greater reductions in inequality.

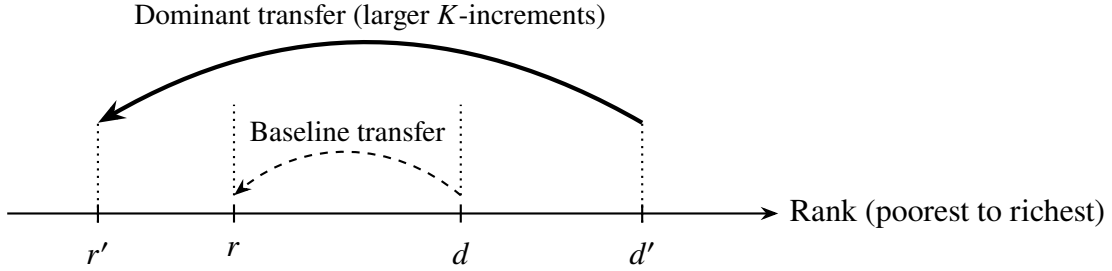


Figure 1: Visualizing how K -dominance captures both the intensity and the reach of progressive transfers. A wider or larger transfer dominates the baseline transfer. The diagram captures the absolute dimension ($K = A$) and the core zero-sum redistribution of the relative dimension ($K = R$).

Recovering the static transfer principles. To see how **K -DPDP** acts as a dynamic extension of the Pigou–Dalton principle, we observe how it evaluates static states in the absolute case. We first formally define the static relation induced by our dynamic framework.

Definition 4 (induced static relation). For any dynamic inequality pre-order $\succsim$ defined on the transition space $\Omega_{(n,\mu)} \times \Omega_{(n,\mu)}$, its *induced static binary relation* $\succsim^*$ on $\Omega_{(n,\mu)}$ is defined by evaluating an active reform against the null transition, or status quo:

$$\hat{\mathbf{y}} \succsim^* \hat{\mathbf{x}} \Leftrightarrow (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \succsim (\hat{\mathbf{x}}, \hat{\mathbf{x}}).$$

Throughout, $\succ^*$ denotes the asymmetric part of $\succsim^*$.

Next, we formalize the classical axiom of static inequality measurement.

Classical Pigou–Dalton Principle (PD). A static inequality pre-order $\succsim^*$ defined on $\Omega_{(n,\mu)}$ satisfies **PD** if, for any rank-ordered distributions $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n,\mu)}$, we have $\mathbf{y} \succ^* \mathbf{x}$ whenever $\mathbf{y}$ is obtained from $\mathbf{x}$ via a single progressive elementary transfer.

A -DPDP encompasses this classical static axiom. We verify both requirements of the asymmetric relation $\mathbf{y} \succ^* \mathbf{x}$. Suppose $\mathbf{y}$ is obtained from $\mathbf{x}$ via a progressive elementary transfer. By **Proposition 1**, the absolute increment $\mathbf{y} - \mathbf{x}$ decomposes into progressive increment transfers, meaning it A -dominates $\mathbf{0}$. **A -DPDP** therefore dictates $(\mathbf{x}, \mathbf{y}) \succ (\mathbf{x}, \mathbf{x})$, establishing the weak preference $\mathbf{y} \succsim^* \mathbf{x}$. Moreover, the reverse increment $\mathbf{x} - \mathbf{y}$ is regressive, meaning $\mathbf{0}$ A -dominates it. **A -DPDP** forces $(\mathbf{y}, \mathbf{y}) \succ (\mathbf{y}, \mathbf{x})$, confirming that maintaining the new baseline is better than reverting the reform. This precludes the

reverse static preference $\mathbf{x} \succ^* \mathbf{y}$. Together, these dual dynamic evaluations imply $\mathbf{y} \succ^* \mathbf{x}$, proving that any dynamic pre-order satisfying **A-DPDP** satisfies classical **PD**.

For the relative dimension ($K = R$), evaluating the dynamic framework against the null transition induces a scale-neutral proportional transfer principle. Because a zero-sum exchange of logarithmic income weakly reduces raw-dollar total income, remaining in the constant-mean domain $\mathcal{D}_{(n,\mu)}^+$ requires an offset. The relative framework evaluates this transition as a progressive zero-sum redistribution in log-income space, combined with a uniform, inequality-neutral expansion that restores the prescribed arithmetic mean of income.

Centered Proportional Transfer Principle (CPTP). A static inequality pre-order $\succ^*$ defined on $\Omega_{(n,\mu)}^+$ satisfies **CPTP** if, for any rank-ordered distributions $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n,\mu)}^+$, we have $\mathbf{y} \succ^* \mathbf{x}$ whenever $\mathbf{y}$ is obtained from $\mathbf{x}$ via a single progressive elementary zero-sum exchange of logarithmic income, combined with an inequality-neutral uniform proportional scaling.

CPTP differs from the proportional ex post transfer principle of [Fleurbaey and Michel \(2001\)](#). It consists of a zero-sum progressive Pigou–Dalton transfer in log-income space followed by a common proportional rescaling that restores the prescribed arithmetic mean. The rescaling leaves centered log incomes unchanged and is therefore inequality-neutral under the relative criterion.

By the same forward-and-reversal logic used in the absolute dimension, **R-DPDP** implies that a valid **CPTP** step from $\mathbf{x}$ to $\mathbf{y}$ satisfies $(\mathbf{x}, \mathbf{y}) \succ (\mathbf{x}, \mathbf{x})$, while its regressive reversal satisfies $(\mathbf{y}, \mathbf{y}) \succ (\mathbf{y}, \mathbf{x})$. The first comparison gives $\mathbf{y} \succ^* \mathbf{x}$, and the second rules out $\mathbf{x} \succ^* \mathbf{y}$. Hence $\mathbf{y} \succ^* \mathbf{x}$. Consequently, any dynamic pre-order satisfying **R-DPDP** induces a static pre-order satisfying **CPTP**.

CPTP evaluates progressivity in proportional rather than physical units. Consequently, a **CPTP** improvement need not be Lorenz-improving in levels. The relation between the two criteria is examined in [Section 2.6.3](#).

Baseline Independence. **K-DPDP** governs progressivity by embedding a directional baseline independence: an enhancement improves the evaluation regardless of the status quo. However, it does not dictate how to weight those improvements across different societies. To compare the *magnitude* of reforms across varying baselines under cross-sectional neutrality, we introduce a secondary, equivalence-based independence axiom.

Centered K -Baseline Independence (K -CBI). For $K \in \{A, R\}$, a dynamic inequality pre-order $\succ$ satisfies **K-CBI** if, for any two rank-ordered transitions $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{u}, \mathbf{v})$ whose distributions belong to the relevant domain, their centered K -increment vectors being identical,

$$\tau^K(\mathbf{x}, \mathbf{y}) - \bar{\tau}^K(\mathbf{x}, \mathbf{y})\mathbf{1} = \tau^K(\mathbf{u}, \mathbf{v}) - \bar{\tau}^K(\mathbf{u}, \mathbf{v})\mathbf{1},$$

where $\bar{\tau}^K$ represents the arithmetic mean of the respective K -increment vector, implies the pre-order is indifferent between them: $(\mathbf{x}, \mathbf{y}) \sim (\mathbf{u}, \mathbf{v})$.

K-CBI restricts dynamic evaluations to increment profiles, abstracting from baseline inequality and uniform growth.

For the absolute dimension ($K = A$), because total income is fixed on this domain, the mean absolute increment is zero ($\bar{\tau}^A = 0$). Thus, **A-CBI** collapses to independence over raw increments. It requires that a zero-sum exchange—such as transferring \$10 from the richest to the poorest—yield the same dynamic evaluation whether it occurs in a highly egalitarian society or an unequal one.

For the relative dimension ($K = R$), increments are logarithmic. Centering the log-increment vector isolates pure proportional redistribution from aggregate growth. To understand this principle, consider two ways to compress the proportional gap between a rich and poor individual by a factor of four. Policy A quadruples the poor person's income while leaving the rich unchanged. Policy B achieves the same relative compression via a zero-sum exchange in log-space, doubling the poor person's income and halving the rich person's income. Because uniform proportional growth does not alter relative inequality, both policies exert an identical equalizing force. Subtracting the mean log-growth (the aggregate expansion in Policy A versus the neutral scale in Policy B) reduces both policies to the same centered log-profile. **R-CBI** therefore declares them equally equalizing.

By isolating structural changes from macroeconomic scale, this axiom accommodates cardinal difference-based frameworks like that of **Bossert and Dutta (2019)**. Furthermore, as established in **Section 3**, **K-CBI** encapsulates both Dynamic Translation Invariance (**DTI**) and Dynamic Scale Invariance (**DSI**). While mathematically elegant, the baseline neutrality underlying the **K-CBI** axiom imposes substantive normative restrictions, which we critically examine in **Section 2.6.6**.

2.6 Dynamic Lorenz dominance

The **K-DPDP** and **K-CBI** axioms define the normative rules for equalization. To operationalize them, we introduce a criterion that aggregates rank-based increments into a tractable dominance relation.

We evaluate the cumulative K -increments accruing to the bottom of the distribution. For any transition $(\mathbf{x}, \mathbf{y})$ —restricted to strictly positive incomes when $K = R$ —we compute the partial sums for each $k \in \{1, \dots, n\}$:

For the absolute case ($K = A$):

$$d_k^A(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^k (y_i - x_i).$$

For the relative case ($K = R$):

$$d_k^R(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^k (\ln y_i - \ln x_i).$$

A transition dominates if it advantages every bottom subset of the population, ensuring the poorest k individuals capture at least their population-share (k/n) of any aggregate growth.

Definition 5 (*K-Dynamic Lorenz dominance*). We say that sequence $(\mathbf{x}', \mathbf{y}')$ *K-dynamically Lorenz dominates* sequence $(\mathbf{x}, \mathbf{y})$, written $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, if the primed sequence satisfies the following centered cumulative condition for every $k \in \{1, \dots, n\}$:

$$d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y}) \geq \frac{k}{n} [d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})].$$

To see how **Definition 5** operates, consider its behavior across the two dimensions.

In the absolute dimension ($K = A$), total physical income is fixed, meaning the aggregate absolute increment is zero ($d_n^A = 0$). The centered constraint therefore collapses to a standard bottom-up dominance check:

$$d_k^A(\mathbf{x}', \mathbf{y}') \geq d_k^A(\mathbf{x}, \mathbf{y}).$$

In the relative dimension ($K = R$), progressive transfers inflate the aggregate log-increment ($d_n^R > 0$). Letting $\Delta d_k^K = d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y})$ denote the net advantage at rank k , the criterion demands:

$$\Delta d_k^R \geq \frac{k}{n} \Delta d_n^R, \quad \forall k \in \{1, \dots, n\}.$$

This serves as a dynamic fair-share constraint. It dictates that if a reform expands the aggregate log-pie, the poorest k individuals must capture at least their population-share (k/n) of that expansion.

Recall that we currently assume all distributions share the same total income and population size. The following lemma confirms that $\succsim_{DL}^K$ is a well-behaved pre-order satisfying our core axioms.

Lemma 1. *The K -dynamic Lorenz dominance relation $\succsim_{DL}^K$ is reflexive and transitive and satisfies **S**, **K-DPDP**, and **K-CBI**.*

Proof. For $\mathbf{z} \in \mathbb{R}^n$, define

$$L_k(\mathbf{z}) = \sum_{i=1}^k z_i - \frac{k}{n} \sum_{i=1}^n z_i.$$

Writing $\Delta = \boldsymbol{\tau}'^K - \boldsymbol{\tau}^K$, the definition of K -dynamic Lorenz dominance is equivalently $(\mathbf{x}', \mathbf{y}') \succsim_{DL}^K (\mathbf{x}, \mathbf{y})$ if and only if $L_k(\Delta) \geq 0$ for every k . Reflexivity is immediate. Transitivity follows from the linearity of L_k , since the increment difference for a composite comparison is the sum of the two intermediate differences. Symmetry follows because permutations within either period leave the rank-ordered transition, and hence every L_k , unchanged.

To verify **K-DPDP**, suppose $(\mathbf{x}', \mathbf{y}')$ K -dominates $(\mathbf{x}, \mathbf{y})$. Then

$$\Delta = \sum_{m=1}^M a_m (\mathbf{e}_{r_m} - \mathbf{e}_{d_m}) + \begin{cases} \mathbf{0}, & K = A, \\ c\mathbf{1}, & K = R, \end{cases} \quad M \geq 1, \quad a_m > 0, \quad r_m < d_m.$$

For each progressive component $\delta_m = a_m (\mathbf{e}_{r_m} - \mathbf{e}_{d_m})$, $L_k(\delta_m)$ equals a_m if $r_m \leq k < d_m$ and zero otherwise, while $L_k(c\mathbf{1}) = 0$ for every k . By linearity, $L_k(\Delta) \geq 0$ for every k . Moreover, choosing any m gives $L_{r_m}(\Delta) \geq a_m > 0$, so reverse dominance fails at $k = r_m$. Therefore $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, which establishes **K-DPDP**.

Finally, suppose two rank-ordered transitions $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{u}, \mathbf{v})$ have identical centered K -increment profiles. Since

$$L_k(\mathbf{z} + c\mathbf{1}) = L_k(\mathbf{z}), \quad \text{for every } c \in \mathbb{R},$$

this equality implies

$$L_k(\boldsymbol{\tau}^K(\mathbf{x}, \mathbf{y})) = L_k(\boldsymbol{\tau}^K(\mathbf{u}, \mathbf{v})), \quad \text{for every } k.$$

The dynamic Lorenz inequalities therefore hold with equality in both directions, so $(\mathbf{x}, \mathbf{y}) \sim_{DL}^K (\mathbf{u}, \mathbf{v})$. Thus $\succsim_{DL}^K$ satisfies **K-CBI**. $\blacksquare$

Lemma 1 proves that $\succsim_{DL}^K$ satisfies **K-DPDP**. The next lemma establishes the converse representation. Applying the standard adjacent-rank decomposition to the centered difference between two rank-based increment profiles, it shows that dynamic Lorenz dominance is equivalent to decomposability into the progressive enhancements appearing in the axiom.

Lemma 2. *Suppose $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, where all distributions in both transitions belong to the relevant domain $\mathcal{D}_{(n,\mu)}$ for $K = A$, and $\mathcal{D}_{(n,\mu)}^+$ for $K = R$. Let $\boldsymbol{\tau}'^K$ and $\boldsymbol{\tau}^K$ be their respective rank-based K -increment vectors. Then:*

- For $K = A$, the difference vector decomposes on the original n -rank space into a finite, possibly empty sum of progressive increment transfers $\delta_1, \dots, \delta_M$:

$$\tau'^A = \tau^A + \sum_{m=1}^M \delta_m.$$

- For $K = R$, the difference vector decomposes on the original n -rank space into a finite, possibly empty sum of progressive increment log-transfers $\delta_1, \dots, \delta_M$, together with a single inequality-neutral uniform shift $c\mathbf{1}$:

$$\tau'^R = \tau^R + \sum_{m=1}^M \delta_m + c\mathbf{1}.$$

Moreover, if $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, then the progressive-transfer collection is nonempty, $M \geq 1$.

Proof. Define the net difference vector on the original n -rank space by

$$\Delta = \tau'^K - \tau^K.$$

For each $k \in \{1, \dots, n\}$, let

$$P_k = \sum_{i=1}^k \Delta_i$$

denote the running cumulative sum of these differences. By the definition of K -dynamic Lorenz dominance,

$$P_k \geq \frac{k}{n} P_n, \quad \text{for all } k \in \{1, \dots, n\}.$$

We now analyze the two dimensions separately.

Case 1: absolute dimension ($K = A$). Because all distributions in $\mathcal{D}_{(n, \mu)}$ share the same total income, absolute increment vectors have zero total sum. Hence

$$P_n = \sum_{i=1}^n (\tau_i'^A - \tau_i^A) = 0.$$

The centered dominance condition therefore reduces to

$$P_k \geq 0, \quad \text{for all } k \in \{1, \dots, n\}.$$

We use the identity

$$\Delta = \sum_{k=1}^{n-1} P_k (\mathbf{e}_k - \mathbf{e}_{k+1}) + P_n \mathbf{e}_n.$$

Since $P_n = 0$, this becomes

$$\Delta = \sum_{k=1}^{n-1} P_k (\mathbf{e}_k - \mathbf{e}_{k+1}).$$

Every term with $P_k > 0$ is a progressive increment transfer of magnitude P_k from rank $k+1$ to rank k . Omitting zero terms and relabeling the remaining nonzero vectors yields a finite, possibly empty collection $\delta_1, \dots, \delta_M$ such that

$$\tau'^A = \tau^A + \sum_{m=1}^M \delta_m.$$

This establishes the absolute decomposition.

Case 2: Relative dimension ($K = R$). In the relative dimension, P_n represents the net aggregate log-increment gap and need not be zero. Define the mean-growth scalar

$$c = \frac{P_n}{n}.$$

Decompose Δ into a centered component and a uniform shift:

$$\Delta = \Delta^c + c\mathbf{1}, \quad \Delta_i^c = \Delta_i - c.$$

Let

$$P_k^c = \sum_{i=1}^k \Delta_i^c$$

be the cumulative partial sum of the centered vector. Then

$$P_k^c = \sum_{i=1}^k (\Delta_i - c) = P_k - kc = P_k - \frac{k}{n}P_n.$$

By the centered dominance condition,

$$P_k^c = P_k - \frac{k}{n}P_n \geq 0, \quad \text{for all } k.$$

Moreover,

$$P_n^c = P_n - n\left(\frac{P_n}{n}\right) = 0.$$

Applying the same identity to the centered vector Δ^c , we obtain

$$\Delta^c = \sum_{k=1}^{n-1} P_k^c (\mathbf{e}_k - \mathbf{e}_{k+1}) + P_n^c \mathbf{e}_n.$$

Since $P_n^c = 0$, this reduces to

$$\Delta^c = \sum_{k=1}^{n-1} \left(P_k - \frac{k}{n}P_n\right) (\mathbf{e}_k - \mathbf{e}_{k+1}).$$

Recombining the centered component with the uniform shift gives

$$\Delta = \sum_{k=1}^{n-1} \left(P_k - \frac{k}{n}P_n\right) (\mathbf{e}_k - \mathbf{e}_{k+1}) + c\mathbf{1}.$$

Each coefficient

$$P_k - \frac{k}{n}P_n$$

is weakly nonnegative by R -dynamic Lorenz dominance. Therefore, every nonzero term in the summation is an progressive increment log-transfer. The trailing term $c\mathbf{1}$ is precisely the inequality-neutral uniform shift in the definition of R -dominance.

If all centered coefficients are zero, then $M = 0$, and the relative difference consists only of the neutral uniform shift $c\mathbf{1}$. Otherwise, omitting zero terms and relabeling the remaining nonzero

vectors yields a finite collection of progressive increment log-transfers $\delta_1, \dots, \delta_M$ such that

$$\tau'^R = \tau^R + \sum_{m=1}^M \delta_m + c\mathbf{1}.$$

This establishes the relative decomposition.

It remains only to verify the final strictness claim. If the original dominance is strict, then the centered cumulative inequality is strict for at least one rank $k < n$. In the absolute case, this means $P_k > 0$ for some $k < n$, and in the relative case, it means

$$P_k - \frac{k}{n}P_n > 0$$

for some $k < n$. Hence at least one coefficient is strictly positive, so the progressive-transfer collection is nonempty, $M \geq 1$. This completes the proof. ■

We define consistency over the relevant transition space $(\Omega_{(n,\mu)} \times \Omega_{(n,\mu)})$ for $K = A$, and $\Omega_{(n,\mu)}^+ \times \Omega_{(n,\mu)}^+$ for $K = R$). A dynamic inequality pre-order $\succsim$ is *K-DL-consistent* if it preserves the dynamic Lorenz ordering. Specifically, evaluating unordered transitions via their rank-ordered counterparts, we require:

$$(\mathbf{x}', \mathbf{y}') \succsim_{DL}^K (\mathbf{x}, \mathbf{y}) \Rightarrow (\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succsim (\hat{\mathbf{x}}, \hat{\mathbf{y}})$$

and

$$(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y}) \Rightarrow (\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ (\hat{\mathbf{x}}, \hat{\mathbf{y}}),$$

where un-hatted vectors represent the rank-ordered counterparts of their hatted originals. **Theorem 1** establishes that our basic axioms generate these *K-DL-consistent* extensions, cementing $\succsim_{DL}^K$ as the minimal benchmark for our framework.

Theorem 1. *For $K \in \{A, R\}$, a dynamic inequality pre-order $\succsim$ defined on the corresponding transition space $\Omega_{(n,\mu)} \times \Omega_{(n,\mu)}$ for $K = A$, and $\Omega_{(n,\mu)}^+ \times \Omega_{(n,\mu)}^+$ for $K = R$, satisfies **S**, **K-DPDP**, and **K-CBI** if and only if it is *K-DL-consistent*.*

Proof. ($\Leftarrow$) Assume *K-DL-consistency*. By **Lemma 1**, the canonical dynamic Lorenz dominance relation $\succsim_{DL}^K$ satisfies **S**, the **K-DPDP**, and **K-CBI**.

For **S**, if one transition is obtained from another by permissible permutations, **Lemma 1** gives mutual $\succsim_{DL}^K$ -dominance. By *K-DL-consistency*, the same mutual weak comparisons hold under $\succsim$, hence the two transitions are indifferent.

For **K-DPDP**, if one transition *K*-dominates another, **Lemma 1** gives strict $\succ_{DL}^K$ -dominance. By *K-DL-consistency*, the same strict comparison holds under $\succsim$.

For **K-CBI**, if two transitions have identical centered *K*-increment profiles, **Lemma 1** gives indifference under $\succsim_{DL}^K$, equivalently weak dominance in both directions. By *K-DL-consistency*, both weak comparisons hold under $\succsim$, so the transitions are indifferent. Thus $\succsim$ satisfies all three axioms.

($\Rightarrow$) Assume $\succsim$ satisfies **S**, the **K-DPDP**, and **K-CBI**. Suppose $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succsim_{DL}^K (\hat{\mathbf{x}}, \hat{\mathbf{y}})$. By **S**, these arbitrary transitions are indifferent to their rank-ordered counterparts:

$$(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \sim (\mathbf{x}', \mathbf{y}') \quad \text{and} \quad (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \sim (\mathbf{x}, \mathbf{y}).$$

Evaluating these sorted representations, we have

$$(\mathbf{x}', \mathbf{y}') \succsim_{DL}^K (\mathbf{x}, \mathbf{y}).$$

By **Lemma 2**, the gap between the primed transition and the baseline transition decomposes into a finite, possibly empty collection of progressive transfers and, for the relative dimension, an inequality-neutral uniform shift:

$$\tau'^K = \tau^K + \sum_{m=1}^M \delta_m + \begin{cases} \mathbf{0}, & K = A, \\ c\mathbf{1}, & K = R, \end{cases}$$

where each δ_m is a valid elementary progressive enhancement for dimension K .

Case 1: $M = 0$. The collection of progressive transfers is empty. Hence the increment profiles differ at most by an inequality-neutral uniform shift:

$$\tau'^K = \tau^K + \begin{cases} \mathbf{0}, & K = A, \\ c\mathbf{1}, & K = R. \end{cases}$$

Equivalently, their centered increment profiles coincide:

$$\tau'^K - \bar{\tau}'^K \mathbf{1} = \tau^K - \bar{\tau}^K \mathbf{1}.$$

Because **K-CBI** applies to the rank-ordered subspace, $(\mathbf{x}', \mathbf{y}') \sim (\mathbf{x}, \mathbf{y})$. Therefore, $(\mathbf{x}', \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$. By transitivity with **S**, we obtain $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ (\hat{\mathbf{x}}, \hat{\mathbf{y}})$.

Case 2: $M \geq 1$. The decomposition yields a finite nonempty sequence of progressive enhancements. By **Definition 3**, the existence of this nonempty progressive sum, combined with the neutral shift for $K = R$, means the primed transition K -dominates the baseline transition. By **K-DPDP**, this superiority mandates $(\mathbf{x}', \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$. In particular, $(\mathbf{x}', \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$. By transitivity with **S**, this establishes the weak implication $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ (\hat{\mathbf{x}}, \hat{\mathbf{y}})$ in K -DL-consistency.

Finally, suppose $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ_{DL}^K (\hat{\mathbf{x}}, \hat{\mathbf{y}})$. Evaluating the unordered transitions through their rank-ordered counterparts, we also have $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$. By the final statement of **Lemma 2**, strict dynamic Lorenz dominance implies $M \geq 1$. Therefore, the argument in Case 2 applies, and **K-DPDP** yields $(\mathbf{x}', \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$. Since

$$(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \sim (\mathbf{x}', \mathbf{y}') \quad \text{and} \quad (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \sim (\mathbf{x}, \mathbf{y}),$$

transitivity of the pre-order preserves strict preference through these indifferences. Hence $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ (\hat{\mathbf{x}}, \hat{\mathbf{y}})$. This establishes the strict implication in K -DL-consistency and completes the proof. ■

Unlike static Lorenz dominance, which compares states in isolation, the dynamic framework ranks the intensity of inequality changes across transitions. However, when evaluating competing reforms for a single economy, the initial baseline distribution is fixed. In this common-baseline setting, **K-CBI** becomes redundant, as the zero-length path in the proof of **Theorem 1** reduces to the equality of the target distributions.

For the absolute dimension, identical increment vectors over a shared baseline $\mathbf{x}$ yield $y'_i - x_i = y_i - x_i$, which forces $\mathbf{y}' = \mathbf{y}$. For the relative dimension, the zero-length path allows the log-increments to differ by a uniform shift ($\tau'^R = \tau^R + c\mathbf{1}$), implying $y'_i = e^c y_i$. Because both final distributions reside in the fixed-mean domain $\mathcal{D}_{(n,\mu)}^+$, their absolute resource sums are identical ($\sum y'_i = \sum y_i = n\mu$), which restricts the scalar multiplier to $e^c = 1$, implying $c = 0$ and $\mathbf{y}' = \mathbf{y}$. The $M = 0$ case thus reduces to reflexivity alone, removing any need for **K-CBI**.

Corollary 1. *If the domain is restricted to transitions sharing a common unordered baseline distribution $\hat{x}$, a dynamic inequality pre-order $\succsim$ satisfies **S** and the **K-DPDP** if and only if it is K-DL-consistent.*

2.6.1 Axiom independence for **Theorem 1**

The three axioms in **Theorem 1** are logically independent: dropping any one of **S**, **K-DPDP**, or **K-CBI** admits a dynamic inequality pre-order satisfying the other two but failing K-DL-consistency. **Appendix A.2** provides the counterexamples.

2.6.2 Endpoint evaluation versus elementary path reconstruction

A traditional transfer axiom is stated locally: a single progressive transfer reduces inequality, and transitivity extends the comparison to finite sequences. It is therefore natural to ask why **K-DPDP** instead applies directly to finite superimpositions of progressive enhancements.

The distinction reflects the object being evaluated. Dynamic inequality ranks transitions through their endpoint rank-based increment profiles, not through a particular sequence of intermediate states. A state-space formulation that applies physical Pigou–Dalton transfers successively to the final distribution need not preserve this object: re-sorting can cause state-space and increment-profile progressivity to diverge. For example, applying the physical transfer $(2, -2)$ to $x = (4.9, 5.1)$ produces the unsorted vector $(6.9, 3.1)$; after re-sorting, the target is $(3.1, 6.9)$ and the realized rank-based increment is $(-1.8, 1.8)$, which is regressive. Thus a progressive enhancement of an increment profile cannot in general be interpreted as a physical transfer applied to its target state.

An elementary formulation directly in increment space is much closer to **K-DPDP**. It would require that

$$\tau'^K = \tau^K + a(e_r - e_d) + c\mathbf{1}, \quad a > 0, \quad r < d,$$

imply strict preference, with $c = 0$ for $K = A$ and $c \in \mathbb{R}$ for $K = R$. The axiom used here is the finite endpoint closure of this elementary condition.

For $K = A$, the two formulations are essentially equivalent: absolute increments are zero-sum, and the Hardy–Littlewood–Pólya decomposition unfolds progressive dominance into a finite sequence of elementary Pigou–Dalton transfers. For $K = R$, path reconstruction is less direct. Each progressive log-transfer generally changes the arithmetic mean of the associated level-income vector, so remaining in $\mathcal{D}_{(n,\mu)}^+$ requires a uniform mean-restoring log-shift after each step. An elementary formulation would therefore also require a path-admissibility result ensuring that the normalized intermediate targets remain rank ordered.

We do not impose this additional procedural structure. The **K-DPDP** evaluates valid endpoint transitions directly: any finite nonempty superimposition of progressive enhancements, together with the neutral uniform shift in the relative case, improves the transition. This preserves the elementary Pigou–Dalton intuition while making the comparison depend only on the transition profiles being ranked.

2.6.3 Static characterizations

Restricting the dynamic framework to null transitions produces two static criteria. In the absolute dimension, the reduction is classical. Evaluating $\succsim_{DL}^A$ against the null transition $(\mathbf{x}, \mathbf{x})$ yields

$$d_k^A(\mathbf{x}, \mathbf{y}) \geq 0, \quad \text{for all } k,$$

which is standard Lorenz dominance, $\mathbf{y} \succsim_L \mathbf{x}$. Likewise, the null version of **A-DPDP** corresponds to the classical Pigou–Dalton transfer principle.

The relative reduction applies the classical majorization machinery to mean-centered log-income vectors (Hardy et al., 1952; Marshall and Olkin, 1979; Marshall et al., 2011). Aboudi and Thon (2003) study weak supermajorization of raw log incomes. Their cumulative-log criterion is generated by product-preserving multiplicative transfers together with coordinatewise income increases, and is therefore analogous to generalized Lorenz welfare dominance after the transformation $x_i \mapsto \ln x_i$. Fleurbaey and Michel (2001) provide a related welfare foundation by characterizing additive welfare functions satisfying proportional and proportional ex post transfer principles. Their proportional ex post transfer is not itself the product-preserving log-space transfer used below; Aboudi and Thon (2003) establish the limiting relationship between the two operations.

Our static criterion removes the mean-log component before applying majorization. It therefore treats uniform proportional scale as inequality-neutral and isolates the distribution of proportional income positions. On a common-geometric-mean domain, raw logarithmic majorization and the centered criterion coincide, just as generalized Lorenz and Lorenz dominance coincide on a common-arithmetic-mean domain. On the fixed-arithmetic-mean domain considered here, centered log-Lorenz dominance implies the raw logarithmic dominance of Aboudi and Thon (2003), but the converse need not hold. This implication follows because centered log-Lorenz dominance and equality of arithmetic means jointly imply that the dominating distribution has a weakly larger geometric mean; adding the resulting mean-log difference back to the centered cumulative inequalities yields the raw cumulative-log inequalities.

We call the resulting scale-neutral economic order *centered log-Lorenz dominance*. We use “centered log-majorization” only as shorthand for ordinary majorization applied to mean-centered log-income vectors; it does not denote a separate mathematical majorization concept.

Definition 6 (centered log-Lorenz dominance). For any rank-ordered distributions $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n, \mu)}^+$, we say that $\mathbf{y}$ *centered log-Lorenz dominates* $\mathbf{x}$, written $\mathbf{y} \succsim_{CL} \mathbf{x}$, if for every $k \in \{1, \dots, n\}$,

$$\sum_{i=1}^k \ln y_i - \frac{k}{n} \sum_{i=1}^n \ln y_i \geq \sum_{i=1}^k \ln x_i - \frac{k}{n} \sum_{i=1}^n \ln x_i.$$

For arbitrary unordered distributions $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \Omega_{(n, \mu)}^+$, centered log-Lorenz dominance is evaluated exclusively through their rank-ordered counterparts.

For any positive distribution $\mathbf{x}$, define its centered log profile by

$$\ell(\mathbf{x}) = \ln \mathbf{x} - \overline{\ln \mathbf{x}} \mathbf{1} = \ln \left(\frac{\mathbf{x}}{\mu_x^G} \right).$$

Then $\mathbf{y} \succsim_{CL} \mathbf{x}$ if and only if $\ell(\mathbf{y})$ Lorenz-majorizes $\ell(\mathbf{x})$. Hence the underlying majorization geometry is classical (Hardy et al., 1952; Marshall and Olkin, 1979; Marshall et al., 2011). The contribution of the result below is to identify the corresponding ordinal axiom on the fixed-arithmetic-mean income domain and to show that centered log-Lorenz dominance is the weakest static inequality

pre-order satisfying Symmetry and CPTP. Here, “weakest means that every static inequality pre-order satisfying these axioms must preserve all weak and strict comparisons made by centered log-Lorenz dominance, although it may make additional comparisons.

The dynamic framework motivates this criterion through the null-transition interpretation of **R-DPDP**. More importantly, centered log-Lorenz dominance also admits a standalone static characterization: it is the dominance relation generated by **S** and **CPTP**.

To formalize this static characterization, we use the standard notion of consistency. A static pre-order $\succ^*$ is consistent with $\succ_{CL}$ if

$$\hat{y} \succ_{CL} \hat{x} \Rightarrow \hat{y} \succ^* \hat{x} \quad \text{and} \quad \hat{y} \succ_{CL} \hat{x} \Rightarrow \hat{y} \succ^* \hat{x}.$$

Theorem 2 (static characterization of centered log-Lorenz dominance). *A static inequality pre-order $\succ^*$ on the unordered domain $\Omega_{(n,\mu)}^+$ satisfies **S** and **CPTP** if and only if it is consistent with centered log-Lorenz dominance ($\succ_{CL}$).*

Proof. Suppose first that $\succ^*$ is consistent with $\succ_{CL}$. Permutations leave ordered centered log profiles unchanged, so consistency in both directions implies **S**. If y is obtained from x by a **CPTP** step, then, for some $a > 0$, $r < d$, and $c \in \mathbb{R}$,

$$\ln y - \ln x = a(e_r - e_d) + c\mathbf{1}.$$

Centering removes $c\mathbf{1}$, and the remaining progressive log-transfer has nonnegative cumulative sums, strictly positive for $r \leq k < d$. Hence $y \succ_{CL} x$, so consistency implies $y \succ^* x$. Thus $\succ^*$ satisfies **CPTP**.

Conversely, suppose that $\succ^*$ satisfies **S** and **CPTP**, and let $y \succ_{CL} x$. By Symmetry, work with the rank-ordered representatives. Set

$$u = \ell(x) \quad \text{and} \quad v = \ell(y).$$

Both vectors are ordered and have zero sum, and $y \succ_{CL} x$ means that v Lorenz-majorizes u . By the ordered Hardy–Littlewood–Pólya theorem (Hardy et al., 1952; Marshall et al., 2011), there is a finite sequence of ordered zero-sum vectors

$$u = z^0, z^1, \dots, z^M = v$$

in which every nontrivial step is a progressive Pigou–Dalton transfer,

$$z^m - z^{m-1} = a_m(e_{r_m} - e_{d_m}), \quad a_m > 0, \quad r_m < d_m.$$

Map each z^m to the fixed-arithmetic-mean income domain by

$$\Phi_\mu(z^m) = \frac{n\mu e^{z^m}}{\sum_{j=1}^n e^{z_j^m}},$$

where exponentiation is componentwise. Since x and y have arithmetic mean μ , $\Phi_\mu(u) = x$ and $\Phi_\mu(v) = y$. Moreover, each nontrivial step satisfies

$$\ln \Phi_\mu(z^m) - \ln \Phi_\mu(z^{m-1}) = a_m(e_{r_m} - e_{d_m}) + c_m \mathbf{1}$$

for some $c_m \in \mathbb{R}$, so it is a **CPTP** step. Transitivity therefore gives $y \succ^* x$ whenever the centered log-majorization is strict. If it is not strict, mutual centered log-majorization gives $\ell(x) = \ell(y)$; hence y is a common proportional rescaling of x , and their common arithmetic mean forces that factor to equal one. Reflexivity and Symmetry then give $y \succ^* x$. Thus $\succ^*$ is consistent with centered log-Lorenz dominance. ■

Why should an analyst or policymaker care about centered log-Lorenz dominance when standard Lorenz dominance already exists? The geometric machinery of $\succ_{CL}$ implements the normative insights of the **CPTP** discussed earlier.

Recall the constant-mean tax reform from

$$x = (10, 50, 90) \quad \text{to} \quad y = (15, 40, 95).$$

Under standard static Lorenz dominance, the absolute evaluation yields non-comparable distributions: the poorest individual gains physical dollars ($15 > 10$), but the poorest two individuals collectively lose them ($55 < 60$). The Lorenz curves cross, rendering the reform unrankable.

Centered log-Lorenz dominance resolves this ambiguity by shifting the lens from physical dollars to proportional shares. Because transferring wealth from the middle class to the poor expands the total sum of log-incomes, the raw relative increments contain an aggregate growth artifact. By evaluating centered increments, $\succ_{CL}$ factors out this implicit uniform scale and aggregates only the pure relative shifts.

Once this scale is removed, the geometry becomes clear: the poorest individual's proportional surge absorbs the middle-class donor's proportional sacrifice. The cumulative centered log-gains remain strictly positive at every nontrivial rank, so the centered log-Lorenz comparison is unambiguous. Because $\succ_{CL}$ isolates this underlying proportional progressivity, it endorses the reform: what standard Lorenz dominance registers as an ambiguous physical crossing becomes a clean relative equalization.

However, $\succ_{CL}$ is not merely a refinement that resolves absolute ambiguities; it operates as a distinct proportional criterion. Consider instead a purely physical progressive transfer from the rich to the middle class, transitioning from

$$x = (10, 50, 90) \quad \text{to} \quad z = (10, 60, 80).$$

Standard Lorenz dominance unequivocally endorses this as an improvement.

Yet, under centered log-Lorenz dominance, the evaluation yields non-comparable distributions. Because the physical transfer expands the aggregate sum of log-incomes, the macroeconomic proportional scale inflates. If the poorest individual's physical income remains fixed at 10, their centered log-position falls. At the same time, the middle-class recipient gains enough proportional ground that the centered cumulative log-gain of the bottom two ranks is positive. The centered log-Lorenz conditions therefore cross, rendering the physical Pigou–Dalton transfer unranked. Thus $\succ_{CL}$ is not merely a sharper version of standard Lorenz dominance; it formalizes a distinct proportional equality paradigm.

2.6.4 Transfer sensitivity, reach, and proportional dominance

Our analysis is related to the transfer-sensitivity criterion of **Shorrocks and Foster (1987)**, but its relation to Pigou–Dalton is different. Shorrocks and Foster strengthen the static absolute criterion by giving greater weight to physical transfers occurring lower in the income distribution. In particular, their favourable composite transfer combines a progressive transfer lower in the distribution with a regressive transfer higher up, under a variance-preserving restriction. Their criterion is therefore position-sensitive and can rank some comparisons for which standard Lorenz curves cross.

The absolute dynamic criterion evaluates a different object. The **A-DPDP** compares rank-based increment profiles across transitions and is sensitive to the amount and *reach* of their progressive content: an enhancement is stronger when it transfers a larger increment or spans a wider rank

interval. Thus, Shorrocks and Foster refine the static Pigou–Dalton/Lorenz criterion through sensitivity to the income position of physical transfers, whereas A-DPDP extends Pigou–Dalton to transition comparisons through sensitivity to the magnitude and rank reach of increment profiles.

The relative criterion differs more fundamentally. Centered log-Lorenz dominance evaluates proportional rather than physical changes after removing uniform proportional scale. It is therefore not a transfer-sensitive refinement of standard Lorenz dominance: a physical Pigou–Dalton improvement may remain unranked by $\succ_{CL}$, while a reform unranked by standard Lorenz dominance may be ranked by $\succ_{CL}$. The criteria thus encode different notions of transfer intensity: Shorrocks–Foster sensitivity is absolute and positional, whereas centered log-Lorenz dominance is proportional and rank-cumulative.

2.6.5 Connection to the cardinal measurement of welfare change

The characterization in [Theorem 1](#) is closely related to the cardinal framework of [Bossert and Dutta \(2019\)](#). They study welfare change under welfare-difference compatibility,

$$V(\mathbf{x}, \mathbf{y}) = W(\mathbf{y}) - W(\mathbf{x}),$$

and characterize generalized-Gini welfare-change measures. For equal population sizes, their associated dominance criterion compares cumulative rank-ordered income differences:

$$\sum_{i=1}^k (y'_i - x'_i) \geq \sum_{i=1}^k (y_i - x_i), \quad k \in \{1, \dots, n\}.$$

On a fixed-mean domain, this condition coincides with $\succ_{DL}^A$, since aggregate absolute increments vanish. Thus, the two approaches share the same fixed-mean absolute dominance geometry.

They differ, however, in object and scope. Bossert and Dutta evaluate cardinal welfare change, so aggregate income growth remains part of the comparison when means vary. We evaluate ordinal inequality change and therefore remove aggregate growth before assessing progressivity. In the absolute dimension, this is achieved by centering raw increments, or equivalently by treating uniform dollar growth as neutral under [DTI](#).

The relative dimension creates a further distinction. Bossert and Dutta’s criterion is based on raw income differences, whereas our R -dimension evaluates centered log-increments. It therefore measures proportional progressivity, removes uniform proportional scale, and yields centered log-Lorenz dominance against the null transition. This relative geometry has no counterpart in their raw-difference framework.

Finally, the axiomatic foundations differ even where the absolute dominance relations coincide. Bossert and Dutta derive cardinal generalized-Gini welfare-change measures from welfare-difference compatibility and related conditions. Our characterization instead concerns an ordinal dynamic inequality pre-order and rests on Symmetry, the K -Dynamic Pigou–Dalton Principle, and Centered K -Baseline Independence. Hence the common fixed-mean absolute criterion arises from different primitives: cardinal welfare differences in their framework and ordinal inequality change in ours.

2.6.6 Baseline sensitivity and the limits of independence

The characterization in [Theorem 1](#) is baseline-neutral: transitions with the same centered K -increment profile receive the same evaluation, regardless of initial inequality. This leaves open

whether a given redistributive improvement should be valued more when it occurs against a more unequal baseline.

It is useful to distinguish two forms of sensitivity. At the micro level, the relative dimension already makes a fixed dollar change more consequential at lower incomes by evaluating logarithmic rather than raw increments. It therefore captures proportional sensitivity without selecting a cardinal utility function. At the macro level, however, *K-CBI* deliberately abstracts from baseline inequality: identical centered increment profiles are treated as equally equalizing across initial distributions.

This restriction is substantive rather than necessary for dynamic inequality measurement. A principle of *Baseline-Dependent Progressivity* could instead assign greater value to the same centered redistributive improvement when the baseline is more unequal. Relaxing *K-CBI* in this direction would yield baseline-sensitive dynamic pre-orders or indices beyond the *K*-dynamic Lorenz benchmark.

3 Distributions with fixed population size and varying total income

We now extend the analysis to transitions with fixed population size n , but variable total income. Let $\Omega_{(n)}$ denote the space of unordered distributions with population size n and positive mean, and let $\Omega_{(n)}^+$ denote the restriction to strictly positive incomes. Let $\mathcal{D}_{(n)}$ and $\mathcal{D}_{(n)}^+$ denote their respective rank-ordered subspaces.

Definition 7. For $K \in \{A, R\}$, a *dynamic inequality pre-order* on the corresponding transition space, $\Omega_{(n)} \times \Omega_{(n)}$ for $K = A$, $\Omega_{(n)}^+ \times \Omega_{(n)}^+$ for $K = R$, is a reflexive and transitive binary relation $\succsim$ on that space.

3.1 Invariance and extensions

To compare reforms across environments with different aggregate income levels, the criterion must specify which uniform changes are treated as distributionally neutral. The relevant neutrality depends on the geometry of the dimension. The absolute dimension evaluates raw income increments, so its neutral operation is an additive shift. The relative dimension evaluates logarithmic increments, so its neutral operation is a multiplicative scale change.

For the absolute dimension, we impose translation invariance: dynamic evaluations are unchanged by independent lump-sum additions in either period.

Dynamic Translation Invariance (DTI). A dynamic inequality pre-order $\succsim$ on $\Omega_{(n)} \times \Omega_{(n)}$ satisfies *DTI* if, for all $\hat{x}, \hat{y} \in \Omega_{(n)}$ and all $\alpha, \beta \in \mathbb{R}$ such that the shifted vectors remain in the domain,

$$(\hat{x}, \hat{y}) \sim (\hat{x} + \alpha \mathbf{1}, \hat{y} + \beta \mathbf{1}).$$

DTI makes absolute dynamic evaluations independent of uniform income shifts in either the baseline or final period. Applied to the null transition, it yields the usual static translation neutrality:

$$(\hat{x}, \hat{x}) \sim (\hat{x}, \hat{x} + \alpha \mathbf{1}),$$

whenever the shifted distribution remains admissible.

For the relative dimension, we impose scale invariance: dynamic evaluations are unchanged by independent proportional rescalings in either period.

Dynamic Scale Invariance (DSI). A dynamic inequality pre-order $\succsim$ on $\Omega_{(n)}^+ \times \Omega_{(n)}^+$ satisfies **DSI** if, for all $\hat{x}, \hat{y} \in \Omega_{(n)}^+$ and all $\alpha, \beta > 0$,

$$(\hat{x}, \hat{y}) \sim (\beta\hat{x}, \alpha\hat{y}).$$

Under **DSI**, dynamic relative evaluations are independent of proportional income scaling in either period. Applied to the null transition, it yields standard static scale neutrality:

$$(\hat{x}, \hat{x}) \sim (\hat{x}, \alpha\hat{x}).$$

Remark 2 (the incompatibility of crossed invariances). The pairing of translation invariance with the absolute dimension and scale invariance with the relative dimension is dictated by the geometry of the two evaluations. Imposing the crossed invariances would erase the changes each dimension is designed to measure. In the absolute dimension, scaling incomes by a factor $\alpha > 1$ widens physical dollar gaps; imposing scale invariance would therefore make the absolute criterion blind to such divergence. Conversely, in the relative dimension, adding a uniform physical amount $\alpha > 0$ to all incomes compresses proportional income ratios; imposing translation invariance would force the relative criterion to ignore this proportional equalization.

Proposition 2. Under **S**, **A-CBI** implies **DTI**, and **R-CBI** implies **DSI**.

Proof. We prove the two implications separately. Because **K-CBI** is restricted to the rank-ordered subspace, we use **S** to bridge unordered transitions to their rank-ordered counterparts.

Absolute dimension. Suppose $\succsim$ satisfies **S** and **A-CBI**. Let $(\hat{x}, \hat{y}) \in \Omega_{(n)} \times \Omega_{(n)}$, and let $\alpha, \beta \in \mathbb{R}$ be such that the translated distributions

$$\hat{u} = \hat{x} + \alpha\mathbf{1}, \quad \hat{v} = \hat{y} + \beta\mathbf{1},$$

remain in $\Omega_{(n)}$.

By **S**, these unordered transitions are indifferent to their rank-ordered counterparts:

$$(\hat{x}, \hat{y}) \sim (x, y) \quad \text{and} \quad (\hat{u}, \hat{v}) \sim (u, v).$$

Because the shifts are uniform, the rank-ordered counterparts satisfy

$$u = x + \alpha\mathbf{1}, \quad v = y + \beta\mathbf{1}.$$

Hence the absolute rank-based increment vector of the translated transition is

$$\tau^A(u, v) = (y + \beta\mathbf{1}) - (x + \alpha\mathbf{1}) = \tau^A(x, y) + (\beta - \alpha)\mathbf{1}.$$

Its arithmetic mean is therefore

$$\bar{\tau}^A(u, v) = \bar{\tau}^A(x, y) + (\beta - \alpha).$$

Subtracting the mean gives

$$\begin{aligned} \tau^A(u, v) - \bar{\tau}^A(u, v)\mathbf{1} &= [\tau^A(x, y) + (\beta - \alpha)\mathbf{1}] - [\bar{\tau}^A(x, y) + (\beta - \alpha)]\mathbf{1} \\ &= \tau^A(x, y) - \bar{\tau}^A(x, y)\mathbf{1}. \end{aligned}$$

Thus the ordered original and translated transitions have identical centered absolute increment profiles. By **A-CBI**, $(x, y) \sim (u, v)$. Using transitivity together with the indifferences given by **S**, we obtain $(\hat{x}, \hat{y}) \sim (\hat{u}, \hat{v})$. Since

$$(\hat{u}, \hat{v}) = (\hat{x} + \alpha\mathbf{1}, \hat{y} + \beta\mathbf{1}),$$

this is precisely **DTI**.

Relative dimension. Suppose $\succsim$ satisfies **S** and **R-CBI**. Let $(\hat{x}, \hat{y}) \in \Omega_{(n)}^+ \times \Omega_{(n)}^+$, and let $\alpha, \beta > 0$. Define

$$\hat{u} = \beta \hat{x}, \quad \hat{v} = \alpha \hat{y}.$$

By **S**, these unordered transitions are indifferent to their rank-ordered counterparts:

$$(\hat{x}, \hat{y}) \sim (x, y) \quad \text{and} \quad (\hat{u}, \hat{v}) \sim (u, v).$$

Because multiplication by a positive scalar preserves ranks, the ordered counterparts satisfy

$$u = \beta x, \quad v = \alpha y.$$

The relative rank-based increment vector of the scaled transition is

$$\begin{aligned} \tau^R(u, v) &= \ln(\alpha y) - \ln(\beta x) \\ &= \ln y - \ln x + (\ln \alpha - \ln \beta) \mathbf{1} \\ &= \tau^R(x, y) + (\ln \alpha - \ln \beta) \mathbf{1}. \end{aligned}$$

Its mean is

$$\bar{\tau}^R(u, v) = \bar{\tau}^R(x, y) + (\ln \alpha - \ln \beta).$$

Therefore

$$\tau^R(u, v) - \bar{\tau}^R(u, v) \mathbf{1} = \tau^R(x, y) - \bar{\tau}^R(x, y) \mathbf{1}.$$

Thus the ordered original and scaled transitions have identical centered relative increment profiles. By **R-CBI**, $(x, y) \sim (u, v)$. Using transitivity together with the indifferences given by **S**, we obtain $(\hat{x}, \hat{y}) \sim (\hat{u}, \hat{v})$. Since $(\hat{u}, \hat{v}) = (\beta \hat{x}, \alpha \hat{y})$, this is precisely **DSI**. ■

3.2 Relative and absolute dynamic Lorenz dominance

We now extend the baseline dynamic Lorenz dominance criteria ($\succsim_{DL}^K$) to the variable-mean domain by embedding the appropriate centering into the cumulative gain functions.

Definition 8 (absolute dynamic Lorenz dominance). A transition $(\hat{x}', \hat{y}') \in \Omega_{(n)} \times \Omega_{(n)}$ *absolutely DL-dominates* $(\hat{x}, \hat{y})$, denoted $(\hat{x}', \hat{y}') \succsim_{DL}^A (\hat{x}, \hat{y})$, if their rank-ordered counterparts satisfy, for every $k \in \{1, \dots, n\}$,

$$\sum_{i=1}^k [(y'_i - \mu_{y'}) - (x'_i - \mu_{x'})] \geq \sum_{i=1}^k [(y_i - \mu_y) - (x_i - \mu_x)],$$

where $\mu_z = \frac{1}{n} \sum_{i=1}^n z_i$ is the arithmetic mean of distribution z .

Definition 9 (relative dynamic Lorenz dominance). A transition $(\hat{x}', \hat{y}') \in \Omega_{(n)}^+ \times \Omega_{(n)}^+$ *relatively DL-dominates* $(\hat{x}, \hat{y})$, denoted $(\hat{x}', \hat{y}') \succsim_{DL}^R (\hat{x}, \hat{y})$, if their rank-ordered counterparts satisfy, for every $k \in \{1, \dots, n\}$,

$$\sum_{i=1}^k \left[\ln \left(\frac{y'_i}{\mu_{y'}^G} \right) - \ln \left(\frac{x'_i}{\mu_{x'}^G} \right) \right] \geq \sum_{i=1}^k \left[\ln \left(\frac{y_i}{\mu_y^G} \right) - \ln \left(\frac{x_i}{\mu_x^G} \right) \right],$$

where $\mu_z^G = (\prod_{i=1}^n z_i)^{1/n}$ is the geometric mean of distribution z .

These variable-mean criteria evaluate centered K -increment profiles. In the absolute dimension,

$$(y_i - \mu_y) - (x_i - \mu_x) = (y_i - x_i) - (\mu_y - \mu_x),$$

so the criterion subtracts the arithmetic mean of the raw absolute increment vector. In the relative dimension,

$$\ln \left(\frac{y_i}{\mu_y^G} \right) - \ln \left(\frac{x_i}{\mu_x^G} \right) = (\ln y_i - \ln x_i) - (\ln \mu_y^G - \ln \mu_x^G),$$

and

$$\ln \mu_y^G - \ln \mu_x^G = \frac{1}{n} \sum_{j=1}^n (\ln y_j - \ln x_j),$$

so the criterion subtracts the arithmetic mean of the log-increment vector. Thus both extensions operate on centered rank-based K -increments.

Consequently, the extended criteria preserve **K-CBI** by construction. Together with **S**, this centered baseline independence yields the appropriate dimension-specific invariance established in **Proposition 2**: Dynamic Translation Invariance in the absolute dimension and Dynamic Scale Invariance in the relative dimension.

Lemma 3. *The extended absolute relation $\succsim_{DL}^A$ satisfies **S**, the **A-DPDP**, and **A-CBI**. Symmetrically, the extended relative relation $\succsim_{DL}^R$ satisfies **S**, the **R-DPDP**, and **R-CBI**.*

Proof. For $K \in \{A, R\}$, let $\tau^K(\mathbf{x}, \mathbf{y})$ denote the rank-based K -increment vector of a transition, and define its centered version by

$$\tilde{\tau}^K(\mathbf{x}, \mathbf{y}) = \tau^K(\mathbf{x}, \mathbf{y}) - \bar{\tau}^K(\mathbf{x}, \mathbf{y})\mathbf{1}.$$

The extended variable-mean criteria are the fixed-population dynamic Lorenz criteria applied to these centered increment profiles.

For the absolute dimension,

$$\tilde{\tau}_i^A(\mathbf{x}, \mathbf{y}) = (y_i - x_i) - (\mu_y - \mu_x) = (y_i - \mu_y) - (x_i - \mu_x).$$

For the relative dimension,

$$\tilde{\tau}_i^R(\mathbf{x}, \mathbf{y}) = (\ln y_i - \ln x_i) - \frac{1}{n} \sum_{j=1}^n (\ln y_j - \ln x_j),$$

and, since

$$\ln \mu_z^G = \frac{1}{n} \sum_{j=1}^n \ln z_j,$$

this is equivalently

$$\tilde{\tau}_i^R(\mathbf{x}, \mathbf{y}) = \ln \left(\frac{y_i}{\mu_y^G} \right) - \ln \left(\frac{x_i}{\mu_x^G} \right).$$

Thus both extended dominance relations compare cumulative sums of centered rank-based K -increments.

S. Permuting individuals in either period only permutes the unordered distributions. After passing to rank-ordered counterparts, the centered K -increment profiles, and therefore all cumulative sums used by $\succsim_{DL}^K$, are unchanged. Hence both extended relations satisfy **S**.

K-DPDP. Suppose $(\mathbf{x}', \mathbf{y}')$ K -dominates $(\mathbf{x}, \mathbf{y})$. In the absolute dimension, the increment difference

is a nonempty finite sum of progressive increment transfers,

$$\tau'^A - \tau^A = \sum_{m=1}^M a_m (\mathbf{e}_{r_m} - \mathbf{e}_{d_m}), \quad a_m > 0, \quad r_m < d_m.$$

In the relative dimension, the increment difference has the form

$$\tau'^R - \tau^R = \sum_{m=1}^M a_m (\mathbf{e}_{r_m} - \mathbf{e}_{d_m}) + c\mathbf{1}, \quad a_m > 0, \quad r_m < d_m.$$

Centering leaves every zero-sum progressive component unchanged and removes the uniform term $c\mathbf{1}$. Hence, in either dimension, the centered increment gap is a nonempty finite sum of progressive increment transfers.

For a single component

$$\delta = a(\mathbf{e}_r - \mathbf{e}_d), \quad a > 0, \quad r < d,$$

its cumulative sums satisfy

$$\sum_{i=1}^k \delta_i = \begin{cases} 0, & k < r, \\ a, & r \leq k < d, \\ 0, & k \geq d. \end{cases}$$

Thus each component gives weakly nonnegative cumulative contributions and a strictly positive contribution for at least one nontrivial rank. Summing over the nonempty collection of components, the extended $\succsim_{DL}^K$ condition holds weakly and strictly at some rank. Therefore $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, so the extended relation satisfies the **K-DPDP**.

K-CBI. If two transitions have identical centered rank-based K -increment profiles, then every cumulative sum entering the extended $\succsim_{DL}^K$ criterion is identical for the two transitions. Hence the dominance inequalities hold with equality in both directions, and the two transitions are indifferent under $\succsim_{DL}^K$. This is exactly **K-CBI**. ■

3.3 Characterization theorems on unconstrained domains

By exploiting the invariance properties embedded within baseline independence, we can map any variable-mean transition onto an equivalent fixed-mean transition with normalized mean $\mu^* > 0$. This lets us generalize our core characterization results to the unconstrained variable-mean spaces. Analogous to the fixed-mean setting, a dynamic inequality pre-order on these unconstrained spaces is *K-DL-consistent* if it preserves the extended dynamic Lorenz ordering.

Theorem 3. *A dynamic inequality pre-order $\succsim$ on $\Omega_{(n)} \times \Omega_{(n)}$ satisfies **S**, **A-DPDP**, and **A-CBI** if and only if it is A-DL-consistent.*

Theorem 4. *A dynamic inequality pre-order $\succsim$ on $\Omega_{(n)}^+ \times \Omega_{(n)}^+$ satisfies **S**, **R-DPDP**, and **R-CBI** if and only if it is R-DL-consistent.*

In the interest of space, we provide the explicit proof for the relative dimension (**Theorem 4**) only; the proof for the absolute dimension (**Theorem 3**) follows by an analogous, symmetric reduction using additive translations in place of multiplicative rescalings.

Proof of Theorem 4. ($\Leftarrow$) Assume $\succsim$ is R-DL-consistent. By **Lemma 3**, the extended relative dynamic Lorenz relation $\succsim_{DL}^R$ satisfies **S**, **R-DPDP**, and **R-CBI** on the variable-mean domain. For

any comparison governed by these axioms, the corresponding weak or strict relation is therefore generated by $\succsim_{DL}^R$. Since R -DL-consistency transfers these comparisons to $\succsim$, the pre-order $\succsim$ satisfies **S**, **R-DPDP**, and **R-CBI**.

($\Rightarrow$) Suppose $\succsim$ satisfies **S**, **R-DPDP**, and **R-CBI** on $\Omega_{(n)}^+ \times \Omega_{(n)}^+$. Let $(\hat{x}, \hat{y})$ and $(\hat{x}', \hat{y}')$ be arbitrary transitions in this variable-mean positive domain, and suppose $(\hat{x}', \hat{y}') \succsim_{DL}^R (\hat{x}, \hat{y})$.

Choose any target arithmetic mean $\mu^* > 0$. Define the scaled distributions

$$\hat{u} = \frac{\mu^*}{\mu_{\hat{x}}} \hat{x}, \quad \hat{v} = \frac{\mu^*}{\mu_{\hat{y}}} \hat{y},$$

and analogously

$$\hat{u}' = \frac{\mu^*}{\mu_{\hat{x}'}} \hat{x}', \quad \hat{v}' = \frac{\mu^*}{\mu_{\hat{y}'}} \hat{y}'.$$

Since all distributions are strictly positive, these scaled distributions remain in the positive domain and satisfy $\hat{u}, \hat{v}, \hat{u}', \hat{v}' \in \Omega_{(n, \mu^*)}^+$.

By **Proposition 2**, **S** together with **R-CBI** implies **DSI**. Hence

$$(\hat{x}, \hat{y}) \sim (\hat{u}, \hat{v}) \quad \text{and} \quad (\hat{x}', \hat{y}') \sim (\hat{u}', \hat{v}').$$

Because the extended relative criterion evaluates centered log-increments, it is invariant under independent positive rescalings of the baseline and target distributions. Indeed, scaling the baseline and target distributions adds constants to the corresponding log-incomes, and those constants are removed by centering. Therefore $(\hat{u}', \hat{v}') \succsim_{DL}^R (\hat{u}, \hat{v})$.

On the fixed-mean domain $\Omega_{(n, \mu^*)}^+ \times \Omega_{(n, \mu^*)}^+$, the extended relative criterion coincides with the fixed-mean criterion in **Definition 5**. Since $\succsim$ satisfies the three axioms globally, its restriction to this fixed-mean subspace satisfies them as well. By **Theorem 1**, $(\hat{u}', \hat{v}') \succsim (\hat{u}, \hat{v})$. Using transitivity together with the **DSI** indifferences, we obtain $(\hat{x}', \hat{y}') \succsim (\hat{x}, \hat{y})$.

The strict implication is identical. If $(\hat{x}', \hat{y}') \succ_{DL}^R (\hat{x}, \hat{y})$, then the same scale invariance preserves strict R -DL dominance: $(\hat{u}', \hat{v}') \succ_{DL}^R (\hat{u}, \hat{v})$. By **Theorem 1**, $(\hat{u}', \hat{v}') \succ (\hat{u}, \hat{v})$. Transitivity of the pre-order preserves strict preference through the **DSI** indifferences, so $(\hat{x}', \hat{y}') \succ (\hat{x}, \hat{y})$. Thus $\succsim$ is globally R -DL-consistent. ■

4 Distributions with varying population size and total income

We now extend our analysis to the unrestricted domain where both population size and total income may vary. Let $\Omega = \bigcup_{n \geq 1} \Omega_{(n)}$ denote the space of unordered distributions of arbitrary population size with positive means, and for any distribution $\hat{x} \in \Omega$, let $n_{\hat{x}}$ denote its population size. Let $\Omega^+ = \bigcup_{n \geq 1} \Omega_{(n)}^+$ denote the restriction to strictly positive incomes. The corresponding unrestricted transition spaces are $\Omega \times \Omega$ for the absolute dimension ($K = A$), and $\Omega^+ \times \Omega^+$ for the relative dimension ($K = R$). Let $\mathcal{D}$ and $\mathcal{D}^+$ denote their respective rank-ordered subspaces.

4.1 Population principle and extensions

Given a distribution $\hat{x} = (\hat{x}_1, \dots, \hat{x}_n)$ and integer $m \geq 1$, the m -replica is

$$\hat{x}^{\{m\}} = (\underbrace{\hat{x}_1, \dots, \hat{x}_1}_{m \text{ times}}, \dots, \underbrace{\hat{x}_n, \dots, \hat{x}_n}_{m \text{ times}}).$$

Dynamic Population Principle (DPP). A dynamic inequality pre-order $\succsim$ satisfies **DPP** if, for any transition $(\hat{x}, \hat{y})$ and any replication integers $l, m \geq 1$, we have

$$(\hat{x}, \hat{y}) \sim (\hat{x}^{\{l\}}, \hat{y}^{\{m\}}).$$

The intuition behind **DPP** is that population replication changes neither baseline inequality nor target inequality. Hence a transition should be evaluated identically after replicating either period: $(\hat{x}, \hat{y}) \sim (\hat{x}^{\{l\}}, \hat{y}^{\{m\}})$. Applied to null transitions, this reduces to the standard static population principle.

Definition 10 (unrestricted dynamic Lorenz dominance). Let $(\hat{x}', \hat{y}')$ and $(\hat{x}, \hat{y})$ be transitions of arbitrary population sizes. Let N be a common multiple of their respective populations, and denote their rank-ordered N -replicas by $\tilde{x}, \tilde{y}, \tilde{x}', \tilde{y}'$, where, for example, $\tilde{x} = \mathbf{x}^{\{N/n_x\}}$.

The transition $(\hat{x}', \hat{y}')$ *absolutely DL-dominates* $(\hat{x}, \hat{y})$, denoted $(\hat{x}', \hat{y}') \succsim_{DL}^A (\hat{x}, \hat{y})$, if, for every $k \in \{1, \dots, N\}$,

$$\sum_{i=1}^k [(\tilde{y}'_i - \mu_{\tilde{y}'}) - (\tilde{x}'_i - \mu_{\tilde{x}'})] \geq \sum_{i=1}^k [(\tilde{y}_i - \mu_{\tilde{y}}) - (\tilde{x}_i - \mu_{\tilde{x}})].$$

Similarly, $(\hat{x}', \hat{y}')$ *relatively DL-dominates* $(\hat{x}, \hat{y})$, denoted $(\hat{x}', \hat{y}') \succsim_{DL}^R (\hat{x}, \hat{y})$, if, for every $k \in \{1, \dots, N\}$,

$$\sum_{i=1}^k \left[\ln \left(\frac{\tilde{y}'_i}{\mu_{\tilde{y}'}} \right) - \ln \left(\frac{\tilde{x}'_i}{\mu_{\tilde{x}'}} \right) \right] \geq \sum_{i=1}^k \left[\ln \left(\frac{\tilde{y}_i}{\mu_{\tilde{y}}} \right) - \ln \left(\frac{\tilde{x}_i}{\mu_{\tilde{x}}} \right) \right].$$

We now formally verify that the expanded dominance criteria satisfy the core axioms on the unrestricted domain.

Lemma 4. *The extended absolute relation $\succsim_{DL}^A$ defined on $\Omega \times \Omega$ satisfies **S**, **A-DPDP**, **A-CBI**, and **DPP**. Symmetrically, the extended relative relation $\succsim_{DL}^R$ defined on $\Omega^+ \times \Omega^+$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP**.*

Proof. We prove the claim for the relative relation; the absolute case is identical after replacing centered log-increments with centered absolute increments.

Let the relevant unrestricted transitions be given. Choose a common population size N for all distributions involved and evaluate each transition through its rank-ordered N -replica. By definition, the unrestricted relation $\succsim_{DL}^R$ is then the variable-mean, fixed-population relation from **Section 3** applied to these N -replicas.

S, R-DPDP, and R-CBI. After replication to the common population size N , the unrestricted relation reduces to the fixed-population variable-mean relation on the replicated rank space. Hence **Lemma 3** applies directly. It follows that the unrestricted relative relation satisfies **S**, the **R-DPDP**, and **R-CBI**.

DPP. Let $(\hat{x}, \hat{y})$ be any transition and let $l, m \geq 1$. To compare $(\hat{x}, \hat{y})$ and $(\hat{x}^{\{l\}}, \hat{y}^{\{m\}})$, choose a common population size N for the four distributions $\hat{x}, \hat{y}, \hat{x}^{\{l\}}, \hat{y}^{\{m\}}$. The N -replica of $\hat{x}^{\{l\}}$ coincides with the N -replica of $\hat{x}$: each original coordinate of $\hat{x}$ appears N/n_x times in both replicas. Similarly, the N -replica of $\hat{y}^{\{m\}}$ coincides with the N -replica of $\hat{y}$. Therefore the centered rank-based log-increment profiles entering the unrestricted criterion are identical for the two transitions. Hence

the dominance inequalities hold with equality in both directions: $(\hat{x}, \hat{y}) \sim_{DL}^R (\hat{x}^{\{l\}}, \hat{y}^{\{m\}})$. This is **DPP**.

The absolute relation satisfies the same properties by the identical replication argument, replacing centered log-increments by centered absolute increments. ■

4.2 Characterization theorems on fully unrestricted domains

The notion of consistency can be extended to the unrestricted transition space in the obvious way: a dynamic inequality pre-order $\succsim$ defined on the unrestricted transition space is *K-DL-consistent* if its weak and strict relations mirror those of the unrestricted dynamic Lorenz dominance relation $\succsim_{DL}^K$.

Theorem 5. *A dynamic inequality pre-order $\succsim$ on $\Omega \times \Omega$ satisfies **S**, **A-DPDP**, **A-CBI**, and **DPP** if and only if it is A-DL-consistent.*

Theorem 6. *A dynamic inequality pre-order $\succsim$ on $\Omega^+ \times \Omega^+$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP** if and only if it is R-DL-consistent.*

We provide the explicit proof for the relative dimension (**Theorem 6**). The proof for the absolute dimension (**Theorem 5**) follows a similar logic and is omitted for brevity; it is obtained symmetrically by substituting the multiplicative scale normalization with an additive mean translation, and invoking **DTI** (guaranteed by **A-CBI** and **S** via **Proposition 2**) in place of **DSI**.

Proof of Theorem 6. ($\Leftarrow$) Assume $\succsim$ is R-DL-consistent. By **Lemma 4**, $\succsim_{DL}^R$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP**. Since R-DL-consistency transfers the weak and strict comparisons generated by $\succsim_{DL}^R$ to $\succsim$, the pre-order $\succsim$ satisfies all four axioms.

($\Rightarrow$) Assume $\succsim$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP**. For any two arbitrary unrestricted transitions $(\hat{x}, \hat{y}), (\hat{x}', \hat{y}') \in \Omega^+ \times \Omega^+$, we use a two-step normalization to map them to the fixed-population, fixed-mean domain.

First, let N be a common multiple of the four populations $n_x, n_y, n_{x'}, n_{y'}$. We replicate each distribution to size N , for example $\tilde{x} = \hat{x}^{\{N/n_x\}}$. By **DPP**, the evaluation is invariant to these replications:

$$(\hat{x}, \hat{y}) \sim (\tilde{x}, \tilde{y}) \quad \text{and} \quad (\hat{x}', \hat{y}') \sim (\tilde{x}', \tilde{y}'). \quad (1)$$

Second, select an arbitrary target arithmetic mean $\mu^* > 0$. We scale the expanded distributions to this common mean by defining

$$\bar{z} = \frac{\mu^*}{\mu_{\tilde{z}}} \tilde{z}, \quad \text{for each } \tilde{z} \in \{\tilde{x}, \tilde{y}, \tilde{x}', \tilde{y}'\}.$$

These normalized unordered distributions now share the same population N and identical arithmetic mean μ^* . By **Proposition 2**, **R-CBI** and **S** imply **DSI**. By **DSI**, the evaluation is indifferent to these normalizations:

$$(\tilde{x}, \tilde{y}) \sim (\bar{x}, \bar{y}) \quad \text{and} \quad (\tilde{x}', \tilde{y}') \sim (\bar{x}', \bar{y}'). \quad (2)$$

Now suppose $(\hat{x}', \hat{y}') \succsim_{DL}^R (\hat{x}, \hat{y})$. By the definition of unrestricted R-DL dominance, replication to the common population size N preserves the dominance relation. Moreover, because the extended relative criterion evaluates centered log-increments, positive rescalings of the baseline and target distributions add only uniform constants to the corresponding log-increment profiles, which are removed by centering. Therefore the normalized transitions satisfy $(\bar{x}', \bar{y}') \succsim_{DL}^R (\bar{x}, \bar{y})$ on the fixed-mean, fixed-population domain $\Omega_{(N, \mu^*)}^+ \times \Omega_{(N, \mu^*)}^+$.

Because $\succsim$ satisfies **S**, **R-DPDP**, and **R-CBI** globally, its restriction to this fixed-mean, fixed-population subspace satisfies these axioms as well. By **Theorem 1**, $(\bar{x}', \bar{y}') \succsim (\bar{x}, \bar{y})$. Transitivity applied through the **DPP** indifference in (1) and the **DSI** indifference in (2) gives $(\hat{x}', \hat{y}') \succsim (\hat{x}, \hat{y})$.

It remains to verify the strict part of **R-DL-consistency**. Suppose $(\hat{x}', \hat{y}') \succ_{DL}^R (\hat{x}, \hat{y})$. Replication to a common population size and positive rescaling preserve strict **R-DL** dominance, since the centered cumulative inequalities are preserved and at least one remains strict. Hence $(\bar{x}', \bar{y}') \succ_{DL}^R (\bar{x}, \bar{y})$. By **Theorem 1**, $(\bar{x}', \bar{y}') \succ (\bar{x}, \bar{y})$. Using transitivity through the **DPP** and **DSI** indifference, we obtain $(\hat{x}', \hat{y}') \succ (\hat{x}, \hat{y})$. Thus $\succsim$ is unrestricted **R-DL-consistent**. ■

Remark 3 (axiom independence for the unrestricted characterizations). The logical independence of the basic axioms (**S**, **K-DPDP**, and **K-CBI**) carries forward from the fixed-population domain via canonical fixed-population embeddings. It remains to establish the independent role of **DPP** in **Theorem 5** and **Theorem 6**.

For either dimension $K \in \{A, R\}$, define a dynamic pre-order on the unrestricted transition space by lexicographically refining unrestricted K -dynamic Lorenz dominance with a population-size tie-breaker. Specifically, for a transition $(\hat{x}, \hat{y})$, let

$$p(\hat{x}, \hat{y}) = n_{\hat{x}} + n_{\hat{y}}.$$

Define $(\hat{x}', \hat{y}') \succsim_{-DPP} (\hat{x}, \hat{y})$ if either $(\hat{x}', \hat{y}') \succ_{DL}^K (\hat{x}, \hat{y})$, or

$$(\hat{x}', \hat{y}') \sim_{DL}^K (\hat{x}, \hat{y}) \quad \text{and} \quad p(\hat{x}', \hat{y}') \leq p(\hat{x}, \hat{y}).$$

This relation is reflexive and transitive because it is a lexicographic refinement of the transitive unrestricted $\succsim_{DL}^K$ relation by a real-valued tie-breaker on K -DL-indifference classes.

The construction satisfies **S**, since permutations do not alter either the unrestricted dynamic Lorenz curves or the population sizes. It satisfies **K-DPDP**, because every K -enhancement yields strict K -DL dominance and therefore bypasses the population-size tie-breaker. It also respects **K-CBI** on fixed-dimensional comparisons: whenever two transitions with the same baseline and target population sizes have identical centered rank-based K -increment profiles, they are K -DL-indifferent and have the same value of $p(\hat{x}, \hat{y})$. Hence the tie-breaker holds with equality in both directions.

However, the construction violates **DPP**. For any transition $(\hat{x}, \hat{y})$, choose replication factors ℓ, m such that

$$p(\hat{x}^{\{\ell\}}, \hat{y}^{\{m\}}) > p(\hat{x}, \hat{y}).$$

The original and replicated transitions are K -DL-indifferent under the unrestricted dominance criterion. But the tie-breaker favors the smaller population representation. Therefore $(\hat{x}, \hat{y}) \succ_{-DPP} (\hat{x}^{\{\ell\}}, \hat{y}^{\{m\}})$, contradicting the indifference required by **DPP**.

5 Concluding remarks

This paper develops an ordinal, index-free framework for comparing inequality changes across income transitions. The central object is the rank-based increment profile of a transition. The absolute dimension evaluates raw income increments, while the relative dimension evaluates logarithmic increments. On fixed-mean domains, Symmetry, the K -Dynamic Pigou–Dalton Principle, and Centered K -Baseline Independence characterize K -dynamic Lorenz dominance. The absolute dimension recovers standard Lorenz dominance against the null transition, while the relative dimension yields centered log-Lorenz dominance. The latter applies ordinary majorization to

mean-centered log incomes and is the scale-neutral Lorenz counterpart of the raw logarithmic order studied by [Aboudi and Thon \(2003\)](#). The relative contribution is therefore not the introduction of log-majorization itself, but its ordinal scale-neutral foundation and, centrally, its extension from comparisons of states to comparisons of transition profiles. The framework extends to variable means through dimension-specific invariance conditions and to variable populations through a Dynamic Population Principle.

Several extensions remain open. First, it would be useful to construct cardinal indices representing these ordinal pre-orders. A dynamic inequality index $J(\hat{x}, \hat{y})$ could be normalized by $J(\hat{x}, \hat{x}) = 0$, so that $J(\hat{x}, \hat{y}) > 0$ indicates a reduction in inequality. A natural question is when such an index can be written as a difference of static inequality indices,

$$J(\hat{x}, \hat{y}) = I(\hat{x}) - I(\hat{y}).$$

For the absolute dimension, the Gini difference is a natural candidate. The relative dimension is less direct, since any representing index must incorporate logarithmic increments, centering, and scale normalization.

Second, the baseline neutrality imposed by Centered K -Baseline Independence could be relaxed. The present framework treats identical centered improvements equally, regardless of initial inequality. A principle of Baseline-Dependent Progressivity would instead assign greater value to the same improvement when it occurs against a more unequal baseline. This would generate baseline-weighted dynamic pre-orders and indices.

A third extension concerns the fair-share benchmark in dynamic Lorenz dominance. The present criterion uses k/n , so the bottom k ranks are benchmarked against their population share of aggregate K -growth. More generally, k/n can be replaced by the cumulative share assigned to those ranks by an exogenous normative weight vector. Non-uniform weights would therefore encode alternative views about how aggregate growth should be allocated across ranks before residual progressivity is assessed. The uniform rule remains the natural benchmark here because it corresponds to the dimension-specific neutrality obtained by centering increments.

Finally, the population extension follows the classical Daltonian replication principle: population replications leave inequality unchanged. Replication preserves relative quantile locations and hence the cumulative rank shares used by dynamic Lorenz dominance, but abstracts from absolute demographic scale. As emphasized by [Carbonell-Nicolau \(2026\)](#), replication can dilute the concentration of fixed elite groups while multiplying the absolute mass of individuals at the bottom of the distribution. If these demographic-scale effects are normatively relevant, the Dynamic Population Principle could be replaced by scale-sensitive population principles that break replication neutrality. Such extensions would allow dynamic inequality comparisons to account for both relative population shares and the absolute number of individuals at different ranks.

A Appendix

A.1 Proof of Proposition 1

Proposition 1. *Any elementary transfer applied to an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ can be decomposed into a finite sequence of globally rank-preserving transfers. Furthermore, if the elementary transfer is progressive (respectively, regressive), its decomposition consists exclusively of globally rank-preserving progressive (respectively, regressive) transfers.*

Proof. We prove the result for a progressive elementary transfer; the regressive case is symmetric. Let the original transfer be from rank d to rank r , with $r < d$, and let its size be $t > 0$. Define, for $\lambda \in [0, t]$,

$$a(\lambda) = x_r + \lambda, \quad b(\lambda) = x_d - \lambda.$$

Since the transfer is elementary and progressive, relative rank preservation gives $a(t) \leq b(t)$. Since $b(\lambda) - a(\lambda)$ is decreasing in λ and is nonnegative at $\lambda = t$, it is nonnegative for every $\lambda \in [0, t]$. Thus $a(\lambda) \leq b(\lambda)$ for all $\lambda \in [0, t]$. Moreover, if equality held at some $\lambda < t$, then the strict decrease of $b(\lambda) - a(\lambda)$ would imply $b(t) - a(t) < 0$, contradicting $a(t) \leq b(t)$. Hence $a(\lambda) < b(\lambda)$ for all $\lambda < t$, with equality possible only at $\lambda = t$.

For each λ , let $z(\lambda)$ be the rank-ordered vector obtained from the multiset

$$\{x_i : i \notin \{r, d\}\} \cup \{a(\lambda), b(\lambda)\}.$$

Thus, $z(0) = \mathbf{x}$ and $z(t) = \mathbf{y}$.

Consider the set of parameter values at which either moving income meets one of the fixed incomes:

$$B = \{0, t\} \cup \{\lambda \in [0, t] : a(\lambda) = x_i \text{ for some } i \notin \{r, d\}\} \\ \cup \{\lambda \in [0, t] : b(\lambda) = x_i \text{ for some } i \notin \{r, d\}\}.$$

This set is finite, since for each fixed income x_i , each of the linear equations $a(\lambda) = x_i$ and $b(\lambda) = x_i$ has at most one solution. List the distinct elements of B in strictly increasing order:

$$0 = \lambda_0 < \lambda_1 < \dots < \lambda_K = t.$$

Fix $k \in \{1, \dots, K\}$. On the open interval $(\lambda_{k-1}, \lambda_k)$, neither $a(\lambda)$ nor $b(\lambda)$ meets any fixed income. Hence their respective ranks in the ordered vector $z(\lambda)$ remain constant on this interval. Let these ranks be p_k and q_k . Since $a(\lambda) < b(\lambda)$ throughout the open interval, the moving recipient value lies strictly below the moving donor value there, and hence $p_k < q_k$.

Set the step size

$$\Delta_k = \lambda_k - \lambda_{k-1} > 0.$$

At either endpoint of this subinterval, one or both moving values may be tied with fixed incomes. In that case, choose an ordering of the tied coordinates by continuity from the open interval $(\lambda_{k-1}, \lambda_k)$: the coordinate occupied by $a(\lambda)$ on the open interval is assigned rank p_k at the adjacent endpoint, and the coordinate occupied by $b(\lambda)$ is assigned rank q_k . Since distributions are evaluated anonymously and ties are rank-indifferent, this representative ordering has no effect on the ordered income vector.

With this representative ordering, the move from $z(\lambda_{k-1})$ to $z(\lambda_k)$ changes only these two coordinates: rank p_k increases by Δ_k , rank q_k decreases by Δ_k , and all other coordinates are unchanged. During the open interval no moving value meets a fixed income, and $a(\lambda) \leq b(\lambda)$ throughout. Hence the intermediate vectors remain weakly ordered. Therefore this step is a globally rank-preserving progressive transfer.

Thus, the original elementary progressive transfer from $\mathbf{x}$ to $\mathbf{y}$ has been decomposed into the finite sequence

$$\mathbf{x} = z(\lambda_0), z(\lambda_1), \dots, z(\lambda_K) = \mathbf{y},$$

where each step is globally rank-preserving and progressive. The regressive case follows by identical logic with the direction of the transfers reversed, completing the proof. ■

A.2 Axiom independence for Theorem 1

We establish independence by constructing, for each axiom, a dynamic pre-order satisfying the other two but failing K -DL-consistency. Let $\hat{z}$ denote an unordered distribution, z its rank-ordered counterpart, and $\hat{\tau}^K$ the unordered K -increment vector.

Dropping S. Define $\succsim_{\neg S}$ on the unordered transition space by

$$(\hat{x}', \hat{y}') \succsim_{\neg S} (\hat{x}, \hat{y}) \Leftrightarrow \begin{cases} (\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y}), \\ \text{or} \\ (\mathbf{x}', \mathbf{y}') \sim_{DL}^K (\mathbf{x}, \mathbf{y}) \text{ and } \hat{\tau}_1'^K - \hat{\tau}_n'^K \geq \hat{\tau}_1^K - \hat{\tau}_n^K. \end{cases}$$

This relation is a reflexive and transitive pre-order: it lexicographically refines K -DL dominance by a real-valued tie-breaker within K -DL-indifference classes, leaving K -DL-incomparable pairs incomparable. It satisfies **K-DPDP** because K -dominance implies strict K -DL dominance. It also satisfies **K-CBI** on the rank-ordered subspace: identical centered K -increment profiles differ at most by a uniform shift, which leaves $\tau_1^K - \tau_n^K$ unchanged.

The relation fails **S**, and hence K -DL-consistency. Take a transition with $\hat{\tau}_1^K - \hat{\tau}_n^K \neq 0$, and obtain $(\hat{x}^\pi, \hat{y}^\pi)$ by swapping individuals 1 and n in both periods. The two transitions have identical rank-ordered counterparts and are therefore K -DL-indifferent, but

$$\hat{\tau}_1^{\pi,K} - \hat{\tau}_n^{\pi,K} = -(\hat{\tau}_1^K - \hat{\tau}_n^K).$$

Thus $\succsim_{\neg S}$ strictly ranks them whenever $\hat{\tau}_1^K - \hat{\tau}_n^K \neq 0$.

Dropping K-DPDP. Define the reverse dynamic Lorenz pre-order by

$$(\hat{x}', \hat{y}') \succsim_{\neg DPDP} (\hat{x}, \hat{y}) \Leftrightarrow d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y}) \leq \frac{k}{n} [d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})] \quad \forall k \in \{1, \dots, n\}.$$

This relation is reflexive and transitive by the same additive argument as K -dynamic Lorenz dominance, with the inequalities reversed. Because it depends only on rank-ordered profiles, it satisfies **S**. Identical centered K -increment profiles satisfy the condition with equality in both directions, so it also satisfies **K-CBI**. However, strict K -DL dominance is ranked in the opposite direction. Hence $\succsim_{\neg DPDP}$ fails K -DL-consistency.

Dropping K-CBI. Let $G(z)$ denote the Gini coefficient of z , and define

$$(\hat{x}', \hat{y}') \succsim_{\neg CBI} (\hat{x}, \hat{y}) \Leftrightarrow \begin{cases} (\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y}), \\ \text{or} \\ (\mathbf{x}', \mathbf{y}') \sim_{DL}^K (\mathbf{x}, \mathbf{y}) \text{ and } G(\mathbf{x}') \leq G(\mathbf{x}). \end{cases}$$

This relation is a reflexive and transitive pre-order: it lexicographically refines K -DL dominance by the baseline Gini coefficient within K -DL-indifference classes. It satisfies **S** because both K -DL dominance and G are permutation invariant, and it satisfies **K-DPDP** because K -dominance implies strict K -DL dominance before the tie-breaker is reached.

It fails **K-CBI**, and hence K -DL-consistency. Choose rank-ordered baselines $\mathbf{x}$ and $\mathbf{u}$ in the relevant domain with $G(\mathbf{u}) > G(\mathbf{x})$. The null transitions $(\mathbf{x}, \mathbf{x})$ and $(\mathbf{u}, \mathbf{u})$ have identical zero

centered K -increment profiles and are therefore K -DL-indifferent, whereas $(x, x) \succ_{\neg\text{CBI}} (u, u)$. Thus none of the three axioms in [Theorem 1](#) follows from the other two.

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